

MHD Derivations for stellar2d — a sympy-verified manuscript

stellar2d development notes
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1 Front matter

1.1 Purpose

This manuscript is an **internal, reproducibility-grade derivation document** for the MHD extension of stellar2d. It covers four parts:

- **Part A** — ideal MHD equations, conservative / primitive forms, flux Jacobian eigensystem, HLLD intermediate states, constrained transport (CT) preservation of $\nabla \cdot \mathbf{B} = 0$; PLM/PPM reconstruction + TVD slope limiters; VL2 predictor-corrector and its CFL bound (hyperbolic + parabolic); HLLD degeneracy branches; Powell-source vs CT comparison; linear-wave convergence IC.
- **Part B** — reduction to a 1D super-radial flux-tube geometry used by the Suzuki-group stellar-wind codes, including the WKB Alfvén-wave action integral, the Parker critical-point condition in this geometry, and the well-balanced MHSE operator needed to keep the atmosphere quiet for long-time wind runs.
- **Part C** — non-ideal dissipation: Ohmic, ambipolar, and their contributions to the energy equation; closure of the two diffusivities through Saha ionisation; sub-grid turbulent heating closure (Suzuki-Inutsuka 2005) for 1D flux-tube wind runs.
- **Part D** — cylindrical shearing-box MHD, the shearing-periodic boundary condition, and the exact Maxwell-Reynolds stress decomposition of the α_{SS} metric.

1.2 Reproducibility protocol

Every algebraic identity in this document is **mechanically verified by sympy**. Each section corresponds to a script in docs/mhd_derivations/scripts/<section>.py that ends with `assert_zero(LHS - RHS, ...)` calls. If any identity cannot be simplified by sympy but is physically correct, we document the alternate verification route (e.g., “manually checked with Wolfram, reason: polynomial simplification blows up”).

The scripts are intentionally **independent**: each one imports only `_common.py` and re-derives

everything it needs from first principles. No cross-script caching of intermediate symbolic results. This makes any section independently re-runnable — a prerequisite for trusting the manuscript as a source of truth for the solver implementation.

1.3 Conventions

- Units are Gaussian with 4π absorbed into $\mathbf{B}$, so that the Alfvén speed is simply $c_A = B/\sqrt{\rho}$, matching Stone & Gardiner 2008, Athena++, PLUTO.
- γ always denotes the ratio of specific heats (not the relativistic Lorentz factor).
- Einstein summation is *not* used; sums are written explicitly.
- sympy variable names mirror the mathematical symbols ($\rho = \rho$, $B_x = B_x$, etc.); see `scripts/_common.py` for the full inventory.

1.4 How to regenerate this manuscript

```
cd docs/mhd_derivations
bash run_all.sh          # (re-)runs every scripts/*.py, refreshes output/
bash build_manuscript.sh # concatenates sections/*.md → manuscript.{md,pdf}
```

If a sympy assertion fails during `run_all.sh`, the build halts and the offending section is flagged. No partial manuscript is emitted.

2 A1. Ideal MHD equations (conservative form)

sympy script: `scripts/a1_ideal_mhd_equations.py` **generated LaTeX:** `output/a1_ideal_mhd_equations.latex.tex` **verifies:** 8 identities (Lorentz $\times$ 3 components, induction $\times$ 3, Poynting-flux identity, $\nabla \cdot (\nabla \times \cdot) = 0$) **code checkpoints (future):** `src/gpu/explicit/athena_mhd_kernels.cu::d_mhd_flux`

2.1 Starting assumptions

Label	Assumption
A1a	Compressible neutral gas with magnetic field $\mathbf{B}$; no viscosity, no Ohmic, no ambipolar, no Hall.
A1b	Infinite conductivity $\rightarrow$ frozen-in flux, $\mathbf{E} = -\mathbf{v} \times \mathbf{B}$.
A1c	Non-relativistic: displacement current $\partial_t \mathbf{E}/c$ dropped.
A1d	Ideal EOS: $p = (\gamma - 1)\rho e$.

2.2 Mass conservation

Written compactly as

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0. \quad (\text{A1-mass})$$

Under A1a–d this is the unmodified fluid continuity equation.

2.3 Lorentz force identity (A1-lorentz)

The single most important vector identity of the derivation is

$$(\nabla \times \mathbf{B}) \times \mathbf{B} = \nabla \cdot (\mathbf{B} \otimes \mathbf{B}) - \nabla \left(\frac{1}{2} |\mathbf{B}|^2 \right) - (\nabla \cdot \mathbf{B}) \mathbf{B}. \quad (\text{A1-lorentz})$$

The last term vanishes **analytically** when the solenoidal constraint $\nabla \cdot \mathbf{B} = 0$ is enforced. At the discrete level it does not vanish in general; this is precisely what CT schemes (§A5) are engineered to handle.

Sympy verification. All three vector components of the residual $(\nabla \times \mathbf{B}) \times \mathbf{B} - [\nabla \cdot (\mathbf{B} \otimes \mathbf{B}) - \nabla(\frac{1}{2}|\mathbf{B}|^2) - (\nabla \cdot \mathbf{B})\mathbf{B}]$ reduce to 0 under `sympy.simplify`, without imposing $\nabla \cdot \mathbf{B} = 0$. This confirms the identity is an *algebraic*, not a physical, fact.

2.4 Momentum conservation (divergence form)

Define the **total magneto-fluid pressure**

$$P^* \equiv p + \frac{1}{2} |\mathbf{B}|^2.$$

Then (A1-lorentz) lets us write momentum conservation in divergence form:

$$\boxed{\partial_t(\rho \mathbf{v}) + \nabla \cdot [\rho \mathbf{v} \otimes \mathbf{v} - \mathbf{B} \otimes \mathbf{B} + P^* \mathbf{I}] = \mathbf{0}.} \quad (\text{A1-mom})$$

Implementation note. $\mathbf{B} \otimes \mathbf{B}$ is *not* symmetric in its indices when read as a flux tensor F_{ij} (it is actually symmetric, but in contrast the induction tensor $v_i B_j - v_j B_i$ is anti-symmetric). In the kernel, $B_i B_j$ for each $(i, j) \in \{(x, x), (x, y), (x, z), (y, y), (y, z), (z, z)\}$ must be computed on the same stencil as $v_i v_j$ — mixing face-averaged $\mathbf{B}$ with cell-centred $\mathbf{B}$ here is the number-one source of initial HLLD bugs.

2.5 Induction equation — two equivalent forms (A1-induction)

Tensor-divergence form (preferred for a Godunov kernel):

$$\partial_t B_i + \partial_j (v_j B_i - v_i B_j) = 0.$$

Curl form (preferred for a CT update):

$$\partial_t \mathbf{B} = \nabla \times (\mathbf{v} \times \mathbf{B}).$$

Sympy verification. For each component $i \in \{x, y, z\}$ we check that

$$\nabla \times (\mathbf{v} \times \mathbf{B}) \Big|_i \equiv -\partial_j (v_j B_i - v_i B_j)$$

holds as a *pure vector-calculus identity* (no $\nabla \cdot \mathbf{B} = 0$, no $\nabla \cdot \mathbf{v} = 0$ invoked). This is the well-known Jacobi-like identity; all three sympy assertions pass.

2.6 Total-energy conservation (A1-energy)

With

$$E \equiv \rho e + \frac{1}{2}\rho|\mathbf{v}|^2 + \frac{1}{2}|\mathbf{B}|^2,$$

the conservative form is

$$\partial_t E + \nabla \cdot [(E + P^*)\mathbf{v} - \mathbf{B}(\mathbf{B} \cdot \mathbf{v})] = 0.$$

The magnetic-Poynting-flux piece $-\mathbf{B}(\mathbf{B} \cdot \mathbf{v})$ and the enthalpy-like piece $P^*\mathbf{v}$ together make the flux manifestly symmetric-hyperbolic.

Sympy-verified identity. The vector identity $\mathbf{B} \times (\mathbf{v} \times \mathbf{B}) = |\mathbf{B}|^2\mathbf{v} - \mathbf{B}(\mathbf{v} \cdot \mathbf{B})$ lets us rewrite the Poynting flux $\mathbf{S} = \frac{1}{2}\mathbf{B} \times (\mathbf{v} \times \mathbf{B})$ in the form used in (A1-energy). `assert_zero` confirms $\nabla \cdot [\mathbf{B} \times (\mathbf{v} \times \mathbf{B}) - (|\mathbf{B}|^2\mathbf{v} - \mathbf{B}(\mathbf{B} \cdot \mathbf{v}))] = 0$ identically.

2.7 Solenoidal constraint preservation (A1-divB)

Taking the divergence of the induction equation in curl form:

$$\partial_t(\nabla \cdot \mathbf{B}) = \nabla \cdot \nabla \times (\mathbf{v} \times \mathbf{B}) = 0,$$

where the last equality is the identity $\nabla \cdot (\nabla \times \cdot) = 0$. **At the continuous level this is automatic**; at the discrete level it is what CT or Dedner-GLM machinery must engineer. See §A5.

Sympy verification. `assert_zero(div_cart(curl_cart(vxB)))` passes for arbitrary smooth $\mathbf{v}, \mathbf{B}$.

2.8 Compact conservative system

Collecting all fluxes, we write the 8-variable conservative MHD system:

$$\partial_t \mathbf{U} + \partial_i \mathbf{F}_i(\mathbf{U}) = \mathbf{0}, \quad \mathbf{U} = (\rho, \rho\mathbf{v}, \mathbf{B}, E)^T.$$

The explicit form of $\mathbf{F}_i$ follows by assembling the four fluxes above. We defer the component-by-component flux Jacobian and its eigensystem to §A3.

2.9 □ Verification checkpoint (to be wired)

Implementation invariants the kernel must satisfy (enforced through `tests/test_athena_mhd_*.cu` in a future PR):

1. **One-cell linearised update reduces to (A1-mass), (A1-mom), (A1-induction-tensor), (A1-energy) exactly** — zero the $\mathbf{B}$ field and verify that $\mathbf{F}_i$ reduces to the Euler flux used in `athena_vl2`.
2. **Solenoidal residual** $(\nabla \cdot \mathbf{B})_{cell}$ from face-centred $\mathbf{B}$ remains at round-off over 100 sound-crossing times on a smooth periodic IC (field-loop advection, §A5).
3. **Energy-momentum consistency** — on a piecewise-constant linear wave IC, total energy updated via (A1-energy) agrees with $\frac{1}{2}\rho|\mathbf{v}|^2 + \frac{1}{2}|\mathbf{B}|^2 + \rho e$ reconstructed from the primitive fields, to round-off per step.

Any failure on (1)–(3) flags the implementation as inconsistent with §A1 and must be resolved before proceeding to §A2.

3 A2. Conservative $\leftrightarrow$ primitive variable transformation

sympy script: `scripts/a2_conservative_primitive.py` **generated LaTeX:** `output/a2_conservative_primitive.latex.tex` **verifies:** $\frac{\partial \mathbf{W}}{\partial \mathbf{U}} \cdot \frac{\partial \mathbf{U}}{\partial \mathbf{W}} = \mathbb{I}_8$ (all 64 entries symbolically checked to zero after subtraction). **code checkpoints (future):** `athena_mhd_kernels.cu::d_prim_from_cons, d_cons_from_prim`.

3.1 Variable inventory

The conservative 8-vector used inside the Godunov flux loop:

$$\mathbf{U} = (\rho, \rho v_x, \rho v_y, \rho v_z, B_x, B_y, B_z, E)^T.$$

The primitive 8-vector used for reconstruction / Riemann solver input:

$$\mathbf{W} = (\rho, v_x, v_y, v_z, B_x, B_y, B_z, p)^T.$$

The total-energy density closes the map through the ideal EOS:

$$E = \frac{p}{\gamma - 1} + \frac{1}{2}\rho|\mathbf{v}|^2 + \frac{1}{2}|\mathbf{B}|^2. \quad (\text{A2-E})$$

3.2 Pressure extraction (primitive from conservative)

Inverting (A2-E) for p gives the pressure-extraction formula the `d_prim_from_cons` kernel must use:

$$p = (\gamma - 1) \left(E - \frac{|\mathbf{m}|^2}{2\rho} - \frac{1}{2}|\mathbf{B}|^2 \right), \quad \mathbf{m} \equiv \rho\mathbf{v}.$$

Implementation gotcha. For low- β , high-Mach flows (MHD blast, Orszag-Tang late time), the hydro term $|\mathbf{m}|^2/(2\rho)$ and the magnetic term $|\mathbf{B}|^2/2$ can both be within machine-precision of the total E . A naïve subtraction can then hand the EOS a negative p . The kernel must either:

1. Accept a positivity fallback (fall back to isothermal primitive extraction and flag the cell),
or
2. Use the dual-energy formulation (evolve an auxiliary internal-energy variable and use it when the primitive subtraction is unreliable).

This is documented in Stone & Gardiner 2008 §4.6 and implemented in Athena++ as the `DUAL_ENERGY` compile-time flag.

3.3 Forward Jacobian $\partial \mathbf{U} / \partial \mathbf{W}$

sympy produces the 8×8 Jacobian explicitly; it is block-sparse with the $\mathbf{B}$ block trivially the identity, the momentum block $\rho \mathbf{I} + \mathbf{v} \otimes (\partial \mathbf{v} / \partial \mathbf{W})$, and the energy row containing $\partial E / \partial \rho$, $\rho \mathbf{v}$, $\mathbf{B}$, $1/(\gamma - 1)$.

3.4 Backward Jacobian $\partial \mathbf{W} / \partial \mathbf{U}$

The inverse map takes the conservative 8-vector to primitives term by term. The non-trivial rows are the velocity-from-momentum and the pressure-from-energy rows:

$$\begin{aligned} \frac{\partial v_i}{\partial (\rho v_j)} &= \frac{\delta_{ij}}{\rho}, & \frac{\partial v_i}{\partial \rho} &= -\frac{v_i}{\rho}, \\ \frac{\partial p}{\partial \rho} &= (\gamma - 1) \frac{|\mathbf{m}|^2}{2\rho^2} = (\gamma - 1) \frac{1}{2} |\mathbf{v}|^2, \\ \frac{\partial p}{\partial (\rho v_i)} &= -(\gamma - 1) v_i, & \frac{\partial p}{\partial B_i} &= -(\gamma - 1) B_i, & \frac{\partial p}{\partial E} &= \gamma - 1. \end{aligned}$$

3.5 Consistency check (sympy)

The script computes $\frac{\partial \mathbf{W}}{\partial \mathbf{U}}(\mathbf{U}(\mathbf{W})) \cdot \frac{\partial \mathbf{U}}{\partial \mathbf{W}}(\mathbf{W})$ after substituting back so that both Jacobians are evaluated at the same primitive state, and verifies

$$\frac{\partial \mathbf{W}}{\partial \mathbf{U}} \cdot \frac{\partial \mathbf{U}}{\partial \mathbf{W}} = \mathbf{I}_8. \quad (\text{A2-inverse})$$

All 64 entries of the residual matrix are `sympy.simplify-reduced` to 0 and pass `assert_zero`.

3.6 □ Verification checkpoint (to be wired)

When the MHD kernel is written, the test

`tests/test_athena_mhd_roundtrip.cu`

should:

1. Seed random primitive 8-vectors $\mathbf{W}$ with $\rho, p > 0$ and $|\mathbf{B}|^2/(2p) < 10^6$ (stay out of the positivity danger zone).
2. Call `d_cons_from_prim` to compute $\mathbf{U}$.
3. Call `d_prim_from_cons` to recover $\mathbf{W}'$.
4. Assert $\|\mathbf{W} - \mathbf{W}'\|_\infty / \|\mathbf{W}\|_\infty < 10 \varepsilon_{\text{mach}}$.

Any regression of (A2-inverse) in the kernel fails this round-trip.

4 A3. Flux Jacobian $\mathbf{A} = \partial \mathbf{F} / \partial \mathbf{U}$ and its 7-wave eigensystem

sympy script: `scripts/a3_flux_jacobian_eigensystem.py` **generated LaTeX:** `output/a3_flux_jacobian_eigensystem.latex.tex` **symbolically verified:** hydro projection (25 entries), discriminant identities I1 & I2. **numerically verified (20 trials $\times$ 3 y):** eigenvalues match closed-form to

$\leq 5 \times 10^{-15}$; $\text{Ar} - \lambda \mathbf{r}$ residuals $\leq 6 \times 10^{-15}$. **code checkpoints (future):**
`athena_mhd_kernels.cu::d_mhd_eigenvalues_primitive, d_mhd_right_eigenvectors.`

4.1 Working frame

The eigenanalysis is done in **primitive form** in 1D along $\hat{x}$. Primitive 7-vector:

$$\mathbf{W} = (\rho, v_x, v_y, v_z, B_y, B_z, p)^T.$$

B_x is **not** an evolution variable in 1D: $\partial_x B_x = \nabla \cdot \mathbf{B} = 0$ forces $B_x = \text{const}$ across any x -interface. The 1D MHD system therefore has **seven** propagating waves, not eight; the eighth mode of the 3D system is the divergence-cleaning wave handled separately by CT (§A5).

4.2 Primitive-form Jacobian A_W

$$\partial_t \mathbf{W} + A_W(\mathbf{W}) \partial_x \mathbf{W} = \mathbf{0},$$

$$A_W = \begin{bmatrix} v_x & \rho & 0 & 0 & 0 & 0 & 0 \\ 0 & v_x & 0 & 0 & B_y/\rho & B_z/\rho & 1/\rho \\ 0 & 0 & v_x & 0 & -B_x/\rho & 0 & 0 \\ 0 & 0 & 0 & v_x & 0 & -B_x/\rho & 0 \\ 0 & B_y & -B_x & 0 & v_x & 0 & 0 \\ 0 & B_z & 0 & -B_x & 0 & v_x & 0 \\ 0 & \gamma p & 0 & 0 & 0 & 0 & v_x \end{bmatrix}. \quad (\text{A3-jacobian})$$

Hydrodynamic sanity check. Setting $B_x = B_y = B_z = 0$ and projecting onto the (ρ, v_x, v_y, v_z, p) subspace recovers the standard 5×5 hydrodynamic primitive Jacobian. symPy verifies all 25 entries of the projection match the hydro reference.

4.3 Characteristic speeds

Define the four reference speeds:

$$c_{s_0}^2 \equiv \gamma p / \rho, \quad c_{Ax}^2 \equiv B_x^2 / \rho, \quad c_{A\perp}^2 \equiv (B_y^2 + B_z^2) / \rho, \quad c_A^2 \equiv c_{Ax}^2 + c_{A\perp}^2.$$

The fast and slow magnetosonic speeds are the two positive roots of

$$c_{f,s}^2 = \frac{1}{2} \left[(c_{s_0}^2 + c_A^2) \pm \sqrt{(c_{s_0}^2 + c_A^2)^2 - 4c_{s_0}^2 c_{Ax}^2} \right]. \quad (\text{A3-cfs})$$

Discriminant identities (HLLD-stable forms). The closed form (A3-cfs) loses ULPs when $c_{A\perp} \rightarrow 0$ or $c_{s_0} \rightarrow 0$. The HLLD kernel uses the two equivalent identities

$$\boxed{c_f^2 + c_s^2 = c_{s_0}^2 + c_A^2, \quad c_f^2 \cdot c_s^2 = c_{s_0}^2 \cdot c_{Ax}^2}, \quad (\text{A3-discriminant})$$

which symPy verifies symbolically (both reduce to 0 under simplify). With these two identities, c_f and c_s can be recovered from $(c_{s_0}^2, c_A^2, c_{Ax}^2)$ without ever forming the root-of-difference $c_A^2 - c_{s_0}^2$.

4.4 The seven wave speeds

$$\{\lambda_k\}_{k=1}^7 = \{v_x - c_f, v_x - c_{Ax}, v_x - c_s, v_x, v_x + c_s, v_x + c_{Ax}, v_x + c_f\}. \quad (\text{A3-wave-speeds})$$

Here $c_{Ax} \equiv B_x / \sqrt{\rho}$ carries the sign of B_x — this convention lets us write the Alfvén speed once without the $s = \text{sign}(B_x)$ factor that appears in Stone+08 eigenvector formulas. The other convention (unsigned $c_{Ax} = |B_x| / \sqrt{\rho} + s$ in every eigenvector entry) is equivalent; we choose signed- c_{Ax} here because it keeps the 7-wave spectrum contiguous for all $\text{sign}(B_x)$.

4.5 Numerical verification of the spectrum

sympy’s `simplify()` does not handle nested radicals $\sqrt{a \pm \sqrt{b}}$ reliably; the fast / slow eigenvector residuals cannot be reduced to zero purely symbolically. This is a known limit — Stone+08 Appendix B, Roe & Balsara 1996, and the Athena++ source all fall back to **numerical random-sample verification** (the “Roe check”) for the eigensystem. We do the same:

1. Draw 20 random physically admissible states $(\rho, p, v_x, B_x, B_y, B_z)$ with $\rho, p > 0$, $B_x \neq 0$, $|B_\perp| \neq 0$;
2. For each of three ratios of specific heats $\gamma \in \{5/3, 7/5, 4/3\}$, diagonalise A_W numerically with `numpy.linalg.eig`;
3. Sort the seven numerical λ_k and compare to the sorted closed-form spectrum (A3-wave-speeds).

Results over 60 trials:

$$\max |\lambda_{\text{closed}} - \lambda_{\text{numerical}}| < 5 \times 10^{-15}, \quad \max \|\mathbf{A}\mathbf{r} - \lambda\mathbf{r}\|_\infty < 6 \times 10^{-15}.$$

The residuals are at the double-precision floor for matrix eigenvalue computation on a well-conditioned 7×7 matrix — indistinguishable from exact.

4.6 Closed-form right-eigenvectors

The closed forms come from Stone & Gardiner 2008 Appendix B Eqs. (B11)–(B13); they are reproduced verbatim in the Athena++ source (`src/eos/adiabatic_mhd.cpp::LRMHDWaves`). We document them here for the kernel implementation; their correctness is pinned to the community verification in Stone+08 Fig. 28–30.

Define the **amplitude-normalisation coefficients**

$$\alpha_f^2 = \frac{c_{s_0}^2 - c_s^2}{c_f^2 - c_s^2}, \quad \alpha_s^2 = \frac{c_f^2 - c_{s_0}^2}{c_f^2 - c_s^2}, \quad \alpha_f^2 + \alpha_s^2 = 1,$$

and the **direction coefficients**

$$\beta_y = \frac{B_y}{\sqrt{B_y^2 + B_z^2}}, \quad \beta_z = \frac{B_z}{\sqrt{B_y^2 + B_z^2}}, \quad \beta_y^2 + \beta_z^2 = 1,$$

and $s = \text{sign}(B_x)$.

4.6.1 Entropy (contact) wave — $\lambda_4 = v_x$

$$\mathbf{r}_{\text{entropy}} = (1, 0, 0, 0, 0, 0, 0)^T.$$

Pure density perturbation, no velocity, pressure, or magnetic-field change. This is literally what “contact discontinuity” means in MHD — it propagates unchanged at the flow speed.

4.6.2 Alfvén waves — $\lambda_{2,6} = v_x \mp c_{Ax}$

$$\mathbf{r}_{A\pm} = (0, 0, \mp s\beta_z, \pm s\beta_y, -\beta_z\sqrt{\rho}, \beta_y\sqrt{\rho}, 0)^T.$$

Transverse velocity and B -field perturbation, no density or pressure change. This is the canonical transverse electromagnetic wave of MHD.

4.6.3 Fast magnetosonic waves — $\lambda_{1,7} = v_x \mp c_f$

$$\mathbf{r}_{f\pm} = (\rho\alpha_f, \pm\alpha_f c_f, \mp s\alpha_s c_s \beta_y, \mp s\alpha_s c_s \beta_z, \alpha_s c_{s0} \sqrt{\rho} \beta_y, \alpha_s c_{s0} \sqrt{\rho} \beta_z, \alpha_f \gamma p)^T.$$

Longitudinal compression plus in-phase perpendicular- B compression.

4.6.4 Slow magnetosonic waves — $\lambda_{3,5} = v_x \mp c_s$

$$\mathbf{r}_{s\pm} = (\rho\alpha_s, \pm\alpha_s c_s, \pm s\alpha_f c_f \beta_y, \pm s\alpha_f c_f \beta_z, -\alpha_f c_{s0} \sqrt{\rho} \beta_y, -\alpha_f c_{s0} \sqrt{\rho} \beta_z, \alpha_s \gamma p)^T.$$

Longitudinal compression plus out-of-phase perpendicular- B compression.

4.7 Degenerate-limit eigenvectors

When $|\mathbf{B}_\perp| = 0$ (pure longitudinal $\mathbf{B}$), the β_y, β_z definitions above divide by zero. Stone+08 Eq. (B17)-(B20) gives replacement eigenvectors for this case. We do not verify these here — they are documented in the Athena++ source and must be implemented as branch-conditionals in the kernel.

4.8 □ Verification checkpoint (to be wired)

The future

tests/test_athena_mhd_eigensystem.cu

must:

1. For 100 random admissible states, evaluate `d_mhd_eigenvalues_primitive` and verify each returned λ_k agrees with (A3-wave-speeds) to 10^{-12} absolute.
2. For each state, evaluate `d_mhd_right_eigenvectors` and verify $\|A_W(\mathbf{W}) \mathbf{r}_k - \lambda_k \mathbf{r}_k\|_\infty < 10^{-12}$ for all $k \in \{1, \dots, 7\}$.
3. Pressure-positivity stress test: states with $\beta_{\text{plasma}} < 0.01$, verify no NaN / no overflow in α_f, α_s .
4. Degenerate-limit coverage: states with $|\mathbf{B}_\perp|/|\mathbf{B}| < 10^{-10}$, verify the degenerate-limit branch is taken and produces non-zero eigenvectors.

This eigenanalysis is the algebraic backbone of HLLD (§A4); any regression here is a direct bug in every Riemann flux computation.

5 A4. HLLD intermediate states (Miyoshi-Kusano 2005)

sympy script: scripts/a4_hlld_intermediate_states.py **verified:** $p_{\text{tot,L}}^* = p_{\text{tot,R}}^*$ symbolically; Rankine-Hugoniot outer-wave jumps numerically (20 trials, max err 3×10^{-15}). **code checkpoints:** athena_mhd_kernels.cu::d_hlld_flux.

5.1 Wave fan

HLLD partitions the Riemann fan between S_L and S_R into four intermediate plateaus separated by S_L^*, S_M, S_R^* — outer fast, Alfvén, contact, Alfvén, outer fast.

5.2 Contact speed (MK Eq. 38)

$$S_M = \frac{(S_R - v_{xR})\rho_R v_{xR} - (S_L - v_{xL})\rho_L v_{xL} - p_{\text{tot,R}}^* + p_{\text{tot,L}}^*}{(S_R - v_{xR})\rho_R - (S_L - v_{xL})\rho_L}.$$

5.3 Star-region state

$\rho_K^* = \rho_K(S_K - v_{xK})/(S_K - S_M)$ (MK Eq. 43); $p_{\text{tot}}^* = p_{\text{tot,L}}^* + \rho_L(S_L - v_{xL})(S_M - v_{xL}) = p_{\text{tot,R}}^* + \rho_R(S_R - v_{xR})(S_M - v_{xR})$ (MK Eq. 41). **Sympy symbolically verifies** the L/R equality.

B_{yK}^*, v_{yK}^* from MK Eqs. 44-47 (see script). Alfvén speeds $S_{L,R}^* = S_M \mp |B_x|/\sqrt{\rho_{L,R}^*}$.

5.4 Critical implementation note

HLLD star state is NOT an EOS state. Do NOT compute $\mathbf{F}_K^*$ via $p = (\gamma - 1)(E^* - \frac{1}{2}\rho|\mathbf{v}|^2 - \frac{1}{2}|\mathbf{B}|^2)$ — it gives the wrong pressure. Use MK Eq. 64 directly with p_{tot}^* from (A4-Ptot-star). Numerical verification confirms: RH residuals are $\mathcal{O}(1)$ with EOS inversion, $\mathcal{O}(10^{-15})$ with direct MK formula.

5.5 □ Verification

tests/test_athena_mhd_hlld.cu — Brio-Wu shock tube match Stone+08 Fig 28 to $L^1 < 2\%$ at $N = 512$.

6 A5. Constrained Transport and discrete $\nabla \cdot \mathbf{B} = 0$

sympy script: scripts/a5_ct_divergence_preservation.py **verified:** CT telescoping identity $(\nabla \cdot \mathbf{B})^{n+1} - (\nabla \cdot \mathbf{B})^n = 0$; Gardiner-Stone 2005 corner-EMF averaging is 2nd-order accurate. **code checkpoints:** athena_mhd_solver.cu::update_face_B_with_emf, tests/test_athena_mhd_field_loop.cu.

6.1 Grid layout

Yee-like staggered: B_x on vertical faces $(i \pm \frac{1}{2}, j)$, B_y on horizontal faces $(i, j \pm \frac{1}{2})$, E_z on corners.

6.2 CT update (Evans-Hawley 1988)

$$B_x^{i\pm 1/2,j,n+1} = B_x^{i\pm 1/2,j,n} - \frac{\Delta t}{\Delta y} (E_z^{i\pm 1/2,j+1/2} - E_z^{i\pm 1/2,j-1/2}),$$

$$B_y^{i,j\pm 1/2,n+1} = B_y^{i,j\pm 1/2,n} + \frac{\Delta t}{\Delta x} (E_z^{i+1/2,j\pm 1/2} - E_z^{i-1/2,j\pm 1/2}).$$

6.3 The central identity

$$(\nabla \cdot \mathbf{B})_{i,j}^{n+1} = (\nabla \cdot \mathbf{B})_{i,j}^n$$

Sympy symbolically verifies this as a pure telescoping identity — the four corner-EMF contributions cancel exactly, regardless of what values the EMFs take. This is why CT is qualitatively different from Dedner GLM: $\nabla \cdot \mathbf{B} = 0$ is an **exact** property, not a truncation error.

6.4 Gardiner-Stone 2005 corner-EMF averaging

$$E_z^{i+1/2,j+1/2} = \frac{1}{4} (E_z^{x,i+1/2,j} + E_z^{x,i+1/2,j+1} + E_z^{y,i,j+1/2} + E_z^{y,i+1,j+1/2}).$$

Sympy-verified via Taylor expansion: leading error is $\frac{h^2}{8}(\partial_x^2 E_z + \partial_y^2 E_z) + \mathcal{O}(h^4)$, no $\mathcal{O}(h)$ term — matches 2nd-order Godunov accuracy.

6.5 □ Verification

tests/test_athena_mhd_field_loop.cu — GS05 field-loop, 10 crossings, lock $\max |\nabla \cdot \mathbf{B}| < 10\varepsilon_{\text{mach}}$ at every step.

7 A6. Reconstruction (PLM/PPM) and TVD slope limiters

sympy script: scripts/a6_reconstruction_tvd.py **verified:** minmod / vL / MC opposite-sign cancellation; Sweby TVD region $0 \leq \varphi(r) \leq \min(2, 2r)$ for $r \in [0, 4]$; PLM reconstruction second-order; PPM 4-point interpolant 4th-order (exact to cubics); PPM parabola $\int_0^1 W d\xi = a_0$. **code checkpoints:** athena_mhd_kernels.cu::d_reconstruct_primitive_plm, athena_mhd_kernels.cu::d_reconstruct_primitive_ppm, tests/test_athena_mhd_linear_wave_convergence.cu.

7.1 Why this section matters

VL2 + HLLD without a **monotonic** reconstruction is 2nd-order but gives spurious oscillations at every shock. The three limiters below are the only ones we need in practice: **minmod** (safest, dispersive), **van Leer** (smooth, slightly sharper), **MC** (most aggressive, Stone+08 default).

A skipped step here directly causes the class of bugs already seen on stellar2d:

- cart_ale2 swept-remap used donor-cell then MUSCL without characteristic projection — caused the periodic-BC drift at P30/P31.
- Stone+08 App. A is explicit: **reconstruct in primitive variables, then project onto characteristic variables** via the eigenvectors of §A3. Skipping the projection converts crisp fast-mode jumps into diffusive blobs.

7.2 Definitions

$$\sigma_{\min\text{mod}}(a, b) = \begin{cases} 0, & \text{sign}(a) \neq \text{sign}(b), \\ \text{sign}(a) \min(|a|, |b|), & \text{else.} \end{cases}$$

$$\sigma_{\text{vL}}(a, b) = \begin{cases} 0, & ab \leq 0, \\ \frac{2ab}{a+b}, & ab > 0. \end{cases}$$

$$\sigma_{\text{MC}}(a, b) = \text{sign}(a) \min\left(2|a|, \frac{|a+b|}{2}, 2|b|\right) \text{ when } ab > 0, \text{ else } 0.$$

7.3 Sweby TVD region (A6-Sweby)

All three limiters satisfy, with $r = b/a$:

$$0 \leq \varphi(r) \leq \min(2, 2r), \quad r \geq 0.$$

Sympy-verified by numerical sweep ($r \in [0, 4]$, 401 samples). Any limiter in this region is TVD for 1D scalar advection with $\text{CFL} \leq 1$ (Harten 1983; LeVeque 2002 §16).

7.4 PLM reconstruction and order

$$W_{i+1/2}^L = W_i + \frac{1}{2}\sigma_i, \quad \sigma_i = \frac{1}{2}(W_{i+1} - W_{i-1}).$$

Leading error $W_{i+1/2}^L - W(x_{i+1/2}) = -\frac{h^2}{8}W''(x_{i+1/2}) + \mathcal{O}(h^3)$, so the reconstruction is **second order** (no $\mathcal{O}(h)$ term). Sympy-verified: no h^0 or h^1 terms in the residual expansion.

7.5 PPM (Colella-Woodward 1984) 4-point interpolant

$$W_{i+1/2} = \frac{7}{12}(W_i + W_{i+1}) - \frac{1}{12}(W_{i-1} + W_{i+2}),$$

applied to the **cell-averaged** values. Sympy-verified: no $\mathcal{O}(h^0..h^3)$ term in the expansion against the smooth reference; leading error is $-\frac{1}{30}h^4W^{(4)}$.

The parabolic form inside a cell (Colella-Woodward 1984 Eq. 1.6),

$$W(\xi) = a_L + \xi[\Delta a + a_6(1 - \xi)], \quad \Delta a = a_R - a_L, \quad a_6 = 6(a_0 - (a_L + a_R)/2),$$

satisfies $\int_0^1 W d\xi = a_0$ **exactly** (sympy-verified): conservation of cell averages is built in.

7.6 Practical recommendation (Stone+08 Appendix A)

1. Reconstruct $(\rho, \mathbf{v}, \mathbf{B}, p)$ in **primitive** variables (not conservative). This keeps positivity of ρ, p easier to enforce.
2. Project onto characteristic variables $\delta W^{(k)} = \ell_k \cdot \delta W$ using the left- eigenvectors of §A3, limit each wave family separately, then project back $\delta W = \sum_k r_k \delta W^{(k)}$.
3. Use MC as the default; switch to van Leer near strong shocks if MC produces staircase artefacts (rare but documented in Stone+08 Fig 28).

7.7 □ Verification checkpoints

- tests/test_athena_mhd_linear_wave_convergence.cu — 3-resolution linear fast-wave advection; L^1 convergence slope in $[1.9, 2.1]$.
- tests/test_athena_mhd_shock_tube.cu — Brio-Wu, MC limiter, no staircase within the fast rarefaction fan.

8 A7. Van-Leer 2 predictor-corrector (Stone-Gardiner 2009)

sympy script: scripts/a7_vl2_predictor_corrector.py **verified:** amplification factor $g(\xi) = 1 - i\nu\xi - \frac{\nu^2}{2}\xi^2 + \mathcal{O}(\xi^3)$ (Lax-Wendroff 2nd-order); $|g|^2 \leq 1$ on $|\nu| \leq 1$; truncation error $\mathcal{O}(\Delta t^3 + \Delta t \Delta x^2)$. **code checkpoints:** athena_mhd_solver.cu::vl2_predictor_step, athena_mhd_solver.cu::vl2_corrector_step.

8.1 The scheme

$$\mathbf{U}^* = \mathbf{U}^n + \frac{\Delta t}{2} \mathcal{L}[\mathbf{U}^n], \quad \mathbf{U}^{n+1} = \mathbf{U}^n + \Delta t \mathcal{L}[\mathbf{U}^*].$$

This is **midpoint-RK2** in time, paired with any semi-discrete operator $\mathcal{L}$ (here: PLM reconstruction + HLLD flux from §A4, plus CT EMF update from §A5).

8.2 Why VL2 over Strang or CTU

Feature	VL2	Strang	CTU
Dimensional coupling	Unsplit, exact	Split, $\mathcal{O}(\Delta t^2)$ error	Unsplit
Auxiliary state	$\mathbf{U}^*$	per-direction sweep	corner-transport cells
CFL limit (unsplit)	≤ 1	≤ 1 in each direction	≤ 1 (multi-D)
Compatibility with CT	Native (Stone-Gardiner 2009)	Requires extra corner EMF gymnastics	Native
Kernel complexity	Low (2 stages)	Low (per-direction)	Higher (CTU corner sweeps)

Stone+08 CTU is more accurate on discontinuities but significantly more complex; VL2 is the default in Athena++ and what we adopt.

8.3 Fourier (von Neumann) analysis on linear advection

With PLM + upwind flux for $a > 0$ on a smooth Fourier mode,

$$g(\xi) = 1 - i\nu\xi - \frac{\nu^2}{2}\xi^2 + \mathcal{O}(\xi^3), \quad \nu \equiv a\Delta t/\Delta x,$$

matching Lax-Wendroff at leading orders. Sympy-verified the three coefficients. Numerical sweep confirms $|g|^2 \leq 1$ for $\nu \in \{0.1, \dots, 1.0\}$; at $\nu = 1.05$ the amplification is $|g|^2 \approx 1.22 > 1$, confirming the CFL limit is **sharp**.

8.4 Truncation error (A7-consistency)

With the central-flux semi-discrete operator $\mathcal{L}[U]_j = -a(U_{j+1} - U_{j-1})/(2h)$ acting on smooth U , after two VL2 stages,

$$\mathbf{U}_j^{n+1} - U(x_j, t + \Delta t) = \mathcal{O}(\Delta t^3 + \Delta t \Delta x^2).$$

Leading error is a purely dispersive $U_{xxx} a(a^2 - 1)/6$ term, vanishing at $\nu = 1$ (exact for advection on-CFL). Sympy expands the predictor/corrector composition and confirms the residual has no $\mathcal{O}(\varepsilon^0 \dots \varepsilon^2)$ terms (with $\varepsilon = \Delta t = h$).

8.5 CFL constraint (1D)

$$\Delta t \leq C_{\text{CFL}} \frac{\Delta x}{\max_{\text{cells}}(|v_x| + c_f)}, \quad C_{\text{CFL}} \leq 1.$$

See §A8 for multidimensional generalisation and parabolic terms.

8.6 Implementation tips (for athena_mhd_solver.cu)

- Save $\mathbf{U}^n$ **before** the predictor so the corrector can rewind if a positivity check fails.
- The predictor need **not** compute the flux to full order — a 1st- order Godunov flux is enough; the corrector is where the 2nd-order accuracy is paid for. Stone+08 explicitly notes this economy.
- For CT, compute **face-centred** E_z at both predictor and corrector; average at the corners using Gardiner-Stone 2005 (§A5).

8.7 □ Verification checkpoints

- tests/test_athena_mhd_linear_wave_convergence.cu — three resolutions, fast wave, expect L^1 convergence slope in $[1.9, 2.1]$.
- tests/test_athena_mhd_vl2_stability.cu — CFL scan $\nu \in \{0.5, 0.9, 0.99\}$ stable; $\nu = 1.05$ must blow up.

9 A8. MHD CFL time-step constraints

sympy script: scripts/a8_mhd_cfl.py **verified:** FTCS diffusion amplification worst-case at $\xi = \pi$ gives $\sigma \leq 1/2$; c_f limits in three degenerate cases ($B_\perp = 0$, $B_x = 0$, $\mathbf{B} = 0$). **code checkpoints:** athena_mhd_kernels.cu::d_mhd_dt_reduction, athena_mhd_solver.cu::compute_dt.

9.1 Hyperbolic CFL (fast wave dominant)

Combining §A3 (7-wave eigensystem) with §A7 ($|\nu| \leq 1$), the unsplit multidimensional bound is

$$\Delta t_{\text{hyp}} \leq C_{\text{CFL}} \left/ \sum_{d=1}^D \max_{\text{cells}} \left(\frac{|v_d| + c_{f,d}}{\Delta x_d} \right) \right., \quad C_{\text{CFL}} \leq 1,$$

where the fast-magnetosonic speed $c_{f,d}$ in direction d is the positive root of the §A3 discriminant.

9.2 Parabolic CFL (Ohmic + ambipolar diffusion)

For the scalar diffusion $\partial_t U = \eta \partial_x^2 U$ with FTCS central space + forward Euler time, the amplification factor is $g(\xi) = 1 - 4\sigma \sin^2(\xi/2)$ with $\sigma = \eta \Delta t / \Delta x^2$. The worst case $\xi = \pi$ gives $g = 1 - 4\sigma$; stability $|g| \leq 1$ requires $\sigma \leq 1/2$ (sympy-verified).

Combining Ohmic + ambipolar as $\eta_{\text{eff}} = \eta_{\Omega} + \eta_{\text{AD}}$:

$$\Delta t_{\text{para}} \leq \frac{1}{2} \min_{\text{cells}} \frac{\Delta x^2}{\eta_{\Omega} + \eta_{\text{AD}}}.$$

9.3 Fast-speed behaviour at degenerate limits (A8-cf-limits)

Sympy-verified:

Limit	c_f^2
$B_{\perp} = 0$ (tangential-free)	$\max(c_{s_0}^2, c_{Ax}^2)$
$B_x = 0$ (perpendicular only)	$c_{s_0}^2 + c_{A\perp}^2$
$\mathbf{B} = 0$ (hydro)	$c_{s_0}^2$

The first two use numerical fall-back in sympy because `sp.Max` does not simplify against the nested radical; 30 random states confirm agreement to machine precision.

9.4 Combined rule

$$\Delta t = \min(\Delta t_{\text{hyp}}, \Delta t_{\text{para}}).$$

9.5 Kernel-form reduction

Implementation-wise, per cell:

$$\left(\frac{1}{\Delta t}\right)_{i,j,k} = \frac{|v_x| + c_f}{\Delta x} + \frac{|v_y| + c_f}{\Delta y} + \frac{|v_z| + c_f}{\Delta z} + 2 \frac{\eta_{\Omega} + \eta_{\text{AD}}}{\min(\Delta x, \Delta y, \Delta z)^2}.$$

Then Δt^{-1} is reduced (max) over the grid and the global Δt is $C_{\text{CFL}} / \Delta t_{\text{max}}^{-1}$.

9.6 Super-time-stepping (optional, future)

When η is large enough that $\Delta t_{\text{para}} \ll \Delta t_{\text{hyp}}$, the RKL2 super-time-stepping scheme (Meyer+12) advances the parabolic part with N sub-steps per hyperbolic step at $\mathcal{O}(N^2)$ CFL relaxation. Not included in the initial `athena_mhd` implementation; reserved for §C5 future extension when Suzuki turbulent heating is active.

9.7 □ Verification checkpoints

- tests/test_athena_mhd_cfl_advection.cu — advection of smooth wave at $C_{\text{CFL}} = 0.95$; no growth. At $C_{\text{CFL}} = 1.05$ the test **must** go unstable within 10 steps (positive control).
- tests/test_athena_mhd_ohm_diffusion.cu — pure Ohmic field-loop decay; Δt_{para} honored; analytic decay rate matched to $< 10^{-5}$.

10 A9. HLLD degenerate branches

sympy script: scripts/a9_hlld_degeneracy.py **verified:** on the Alfvén locus $(S_K - v_{xK})^2 = B_x^2/\rho_K$, numerator of B_{yK}^* vanishes and denominator reduces to $\rho_K(S_K - v_{xK})(v_{xK} - S_M)$; $D_{S_M} = 0$ when both sides of the Riemann interface are identical. **code checkpoints:** athena_mhd_kernels.cu::d_hlld_flux (branch dispatch).

10.1 Three degeneracies of the generic HLLD

The §A4 formulas B_{yK}^*, v_{yK}^* blow up in three cases. The kernel must dispatch on these **before** evaluating the generic formulas.

Branch	Condition	Fallback
D1	$B_x = 0$	HLLC-MHD (Li 2005), 3-wave fan
D2	$(S_K - v_{xK})^2 = B_x^2/\rho_K$	Upstream B_y, v_y, B_z, v_z unchanged
D3	$S_L \rightarrow S_R$ and sides identical	Pure HLL (2-wave average)

10.2 D1 — Zero longitudinal field ($B_x = 0$)

$$B_x = 0 \implies S_L^* = S_R^* = S_M, \mathbf{U}_K^* = \mathbf{U}_K^{**},$$

so the two-plateau star structure collapses to one plateau. The algorithm reduces to **HLLC-MHD** (Li 2005 Appendix A). Recommended implementation: detect $|B_x| < \varepsilon \cdot (c_{s0} + c_{A\perp})\sqrt{\rho}$ and dispatch.

10.3 D2 — Alfvén locus removable singularity

The MK Eq. 44 denominator $\rho_K(S_K - v_{xK})(S_K - S_M) - B_x^2$ vanishes on the locus $(S_K - v_{xK})^2 = B_x^2/\rho_K$, where the upstream state already sits on the Alfvén characteristic.

Sympy-verified along the locus: - Numerator $\rho_K(S_K - v_{xK})^2 - B_x^2 \rightarrow 0$. - Denominator reduces to $\rho_K(S_K - v_{xK})(v_{xK} - S_M)$.

So both vanish, and B_{yK}^* is $0/0$ — *removable*. The physical content is that the transverse field does not jump across an Alfvén wave that coincides with the upstream state.

Kernel regularisation:

$$|\text{den}| < \epsilon\sqrt{\rho_K}|B_x| \implies B_{yK}^* \leftarrow B_{yK}, v_{yK}^* \leftarrow v_{yK} \text{ (and same for } z\text{)}.$$

with $\epsilon \sim 10^{-12}$ in double precision. Falling through to the upstream state is continuous with the generic formula.

10.4 D3 — $S_L \approx S_R$ (vacuum / strongly-aligned)

When $\|(S_R - v_{xR})\rho_R - (S_L - v_{xL})\rho_L\| < \epsilon$, the S_M denominator vanishes. Physically: the Riemann fan has collapsed ($L = R$), so there is no jump to resolve. Fall back to pure HLL:

$$\mathbf{F}_{\text{HLL}} = \frac{S_R \mathbf{F}_L - S_L \mathbf{F}_R + S_L S_R (\mathbf{U}_R - \mathbf{U}_L)}{S_R - S_L}.$$

Pure HLL is diffusive but unconditionally stable, a safe last-resort.

10.5 Implementation order (most to least specific)

```
if |D_SM| < ε:
    return HLL_flux()           # D3
if |B_x| < ε * (cs0 + cA1)*sqrt(ρ):
    return HLLC_MHD_flux()     # D1
compute generic star states (§A4)
if |den_K| < ε * sqrt(ρ_K) * |B_x|: # D2
    set B_y*_K, B_z*_K, v_y*_K, v_z*_K
    to upstream values
return HLLD_flux(generic)
```

10.6 Why this matters

Without these dispatches, the Brio-Wu shock tube **will** produce NaN at $t \sim 0.05$ when a rarefaction fan grazes the Alfvén locus. Documented in Miyoshi-Kusano 2005 Sec. 3.4 and in Athena++ `src/hydro/rsolvers/mhd/hlld.cpp::HLLDTransport` branch logic.

10.7 □ Verification checkpoints

- `tests/test_athena_mhd_brio_wu.cu` — Brio-Wu, 512 cells, match Stone+08 Fig 28 to $L^1 < 2\%$.
- `tests/test_athena_mhd_degenerate_Bx0.cu` — synthetic state with $B_x = 10^{-14}$, must not produce NaN and must match pure-HLLC hydrodynamic flux to 10^{-10} .

11 A10. Powell 8-wave source vs Constrained Transport

sympy script: `scripts/a10_powell_vs_ct.py` **verified:** Powell source term $\mathbf{S}_p$ vanishes identically when $\nabla \cdot \mathbf{B} = 0$; since §A5 gives CT preservation at the discrete level, $\mathbf{S}_p \equiv 0$ for all n and no correction is needed. **code checkpoints:** no new code required — documented to prevent future “add a Powell source term for safety” PR.

11.1 The 8-wave system (Powell+99)

The 8-wave formulation augments ideal MHD with a divergence-cleaning wave and a **non-conservative** source term:

$$\partial_t \mathbf{U} + \partial_i \mathbf{F}_i(\mathbf{U}) = -(\nabla \cdot \mathbf{B}) \mathbf{S}_p(\mathbf{U}),$$

with

$$\mathbf{S}_p = \begin{pmatrix} 0 \\ \mathbf{B} \\ \mathbf{v} \\ \mathbf{B} \cdot \mathbf{v} \end{pmatrix}.$$

At the **continuous** level, $\nabla \cdot \mathbf{B} = 0$ exactly, so $\mathbf{S}_p \cdot 0 = 0$ — trivial.

11.2 The non-trivial claim

At the **discrete** level, a cell-centred or vertex-centred $\mathbf{B}$ storage generally has $(\nabla \cdot \mathbf{B})_{i,j} \neq 0$ at $\mathcal{O}(\Delta x)$, and the Powell source is a genuine correction. **But** on the Yee-staggered grid with CT (§A5), the discrete $(\nabla \cdot \mathbf{B})_{i,j}^n$ satisfies

$$(\nabla \cdot \mathbf{B})_{i,j}^n = (\nabla \cdot \mathbf{B})_{i,j}^0 \quad \forall n$$

(the §A5 telescoping identity). Provided initialisation seeds $\mathbf{B}^0$ from a vector potential or via one projection solve $(\nabla^2 \phi = \nabla \cdot \mathbf{B}_{\text{raw}}, \mathbf{B} \leftarrow \mathbf{B}_{\text{raw}} - \nabla \phi)$, $(\nabla \cdot \mathbf{B})^0 = 0$ and thus

$$\boxed{\mathbf{S}_p^n \equiv 0 \quad \forall n.}$$

11.3 GLM-MHD contrast (Dedner+02)

GLM adds an 8th field ψ and hyperbolic-parabolic cleaning:

$$\partial_t \psi + c_h^2 (\nabla \cdot \mathbf{B}) = -\alpha \psi, \quad \partial_t \mathbf{B} + \nabla \psi = \dots$$

GLM **does not** preserve $\nabla \cdot \mathbf{B} = 0$ exactly; it advects and damps the constraint violation at wave speed c_h . CT is *exact* by construction; GLM trades exactness for a grid that does not need face-centred field storage (ADER-DG codes etc.).

For `athena_mhd` we use CT. This section exists to lock in “no Powell source needed” as a *derived* result, not an assumption.

11.4 Implementation note

The kernel must:

1. Seed $\mathbf{B}^0$ from a vector potential $\mathbf{A}$ via $\mathbf{B} = \nabla \times \mathbf{A}$ at face centres. No projection needed; $\nabla \cdot \mathbf{B} \equiv 0$ by vector calculus.
2. Run the diagnostic `d_mhd_divB_check` after every time step to confirm $\max |\nabla \cdot \mathbf{B}| < 10 \epsilon_{\text{mach}}$. If this check ever fails, the bug is in Step 1 or in the CT kernel; not a reason to add a Powell source.

11.5 □ Verification checkpoints

- `tests/test_athena_mhd_field_loop.cu` — Gardiner-Stone 2005 field loop, 10 diagonal advections. Track $\max |\nabla \cdot \mathbf{B}|$ — must remain at machine precision.
- No test for the Powell source — it is **not** in the kernel and should never be added.

12 A11. Linear MHD wave initial conditions (Stone+08 Tab 1)

sympy script: `scripts/all_linear_wave_initial_conditions.py` **verified:** with $\rho_0 = 1, p_0 = 1/\gamma, \mathbf{B}_0 = (1, \sqrt{2}, 1/2), \gamma = 5/3$ the wave speeds are $c_f = 2, c_s = 1/2, c_{Ax} = 1, c_{s_0} = 1$, all matching Stone+08 Table 1; the discriminant identities $c_f^2 + c_s^2 = c_{s_0}^2 + c_A^2$ and $c_f^2 c_s^2 = c_{s_0}^2 c_{Ax}^2$ are sympy-verified. **code checkpoints:** `tests/test_athena_mhd_linear_wave_convergence.cu; tst/test_athena_mhd/test_linwave.py` (Python harness analogous to the existing `tst/test_ale2/test_linwave.py`).

12.1 Why this is the “gold-standard” convergence test

Linear wave advection has a known exact solution: after one period the perturbation returns to its initial shape, and any deviation is **purely numerical**. The L^1 norm of the difference, sampled at multiple resolutions, directly measures convergence slope and diagnoses accuracy loss of the scheme.

A scheme that claims 2nd order but gives slope 1.3 in the fast-wave test has a broken reconstruction, broken HLLD, or broken CT — and this test finds that before `athena_mhd` touches a shock tube.

12.2 Background (Stone+08 Table 1)

$$\rho_0 = 1, \quad p_0 = \frac{1}{\gamma}, \quad \mathbf{v}_0 = \mathbf{0}, \quad \mathbf{B}_0 = \left(1, \sqrt{2}, \frac{1}{2}\right), \quad \gamma = \frac{5}{3}.$$

Derived wave speeds (all sympy-verified):

$$c_f = 2, \quad c_{Ax} = 1, \quad c_s = \frac{1}{2}, \quad c_{s_0} = 1.$$

12.3 Perturbation

For mode k with eigenvector $\mathbf{r}_k$ (from §A3) and amplitude $A = 10^{-6}$:

$$\mathbf{W}(x, 0) = \mathbf{W}_0 + A \mathbf{r}_k \cos\left(\frac{2\pi x}{L}\right), \quad x \in [0, L].$$

12.4 Periods on an L -periodic box

$$T_f = L/2, \quad T_A = L, \quad T_s = 2L.$$

12.5 Entropy-mode exception

The entropy eigenvalue is $\lambda_{\text{ent}} = v_{0,x}$ which is 0 at our base state. To make the entropy-wave convergence test meaningful, **choose** $v_{0,x} = 1$ **for the entropy-mode case only**; then $T_{\text{ent}} = L$. Stone+08 does the same (Sec 6.1).

12.6 $\nabla \cdot \mathbf{B} = 0$ at IC (automatic)

In the 1D MHD system (§A3) B_x is a parameter, not an evolution variable; all seven right-eigenvectors live in $(\rho, v_x, v_y, v_z, B_y, B_z, p)$ and do not perturb B_x . For a plane wave along $\hat{x}$, $\nabla \cdot \delta \mathbf{B} = i k_x \delta B_x = 0$ automatically. **No extra projection step needed at IC.**

12.7 Expected error

For a VL2 + HLLD + CT code (§A7, §A4, §A5),

$$\varepsilon_{L^1}(k, \Delta x) = C_k A (\Delta x/L)^2,$$

with C_k an $\mathcal{O}(1)$ mode-dependent constant. Convergence slope measured across $\Delta x = L/64, L/128, L/256$ should fall in $[1.9, 2.1]$ for all 7 modes.

12.8 Practical test protocol

1. For each of the 7 modes:
 - Run the solver at resolutions $N \in \{64, 128, 256\}$.
 - Evolve for one period T_k .
 - Compute $\varepsilon_{L^1}(N) = \frac{1}{N} \sum_j |W_j^N - W_j^{\text{exact}}|$.
2. Fit slope: $\log \varepsilon \propto p \log \Delta x$.
3. Accept if $|p - 2| < 0.1$.

12.9 □ Verification checkpoint

tests/test_athena_mhd_linear_wave_convergence.cu — single test that cycles through 7 modes $\times$ 3 resolutions $\times$ 1 period; outputs a 21-row CSV; asserts slope $\in [1.9, 2.1]$ per mode.

12.10 Extension: 3D linear wave

A 3D linear wave IC is constructed identically, substituting $k_x \rightarrow \mathbf{k}$ and rotating the eigenvector by the same angle. Stone+08 §6.2 does the oblique test at $\mathbf{k} = (1, 2, 2)$. This stresses dimensional coupling. Reserved for post-MVP; not included in the initial test suite.

13 B1. Super-radial flux-tube reduction

sympy script: scripts/b1_flux_tube_geometry.py **verified:** flux conservation $AB_r = \text{const}$; MHSE $\partial_r p + \rho g = -B_r^2 \partial_r (\ln A)$; Kopp-Holzer $f(R_*) = 1$, $f(\infty) = f_{\text{max}}$. **code checkpoints:** tests/test_mhd_wind_hse_stationary.cu.

13.1 Geometry

$A(r) = r^2 f(r)$, $f(R_*) = 1$, $f(\infty) = f_{\text{max}}$. Kopp-Holzer form:

$$f(r) = \frac{f_{\text{max}} e^{(r-R_1)/\sigma} + f_1}{e^{(r-R_1)/\sigma} + 1}, \quad f_1 = 1 - (f_{\text{max}} - 1) e^{(R_*-R_1)/\sigma}.$$

13.2 1D flux-tube equations

$$\begin{aligned}\partial_t(\rho A) + \partial_r(\rho v_r A) &= 0, \\ \partial_t(\rho v_r A) + \partial_r[A(\rho v_r^2 + p_{\text{tot}} - B_r^2)] &= (p_{\text{tot}} - B_r^2)\partial_r A - \rho g A, \\ \partial_t(EA) + \partial_r[A((E + p_{\text{tot}})v_r - B_r(\mathbf{B} \cdot \mathbf{v}))] &= -\rho g v_r A + A Q.\end{aligned}$$

13.3 Flux conservation

$$A(r)B_r(r) = R_*^2 B_0 \Rightarrow B_r(r) = \frac{B_0}{f(r)} (R_*/r)^2.$$

13.4 MHSE

$$\partial_r p + \rho g = -B_r^2 \partial_r (\ln A).$$

IC gotcha: for a Suzuki-style wind run, atmospheres must be integrated from this MHSE, NOT plain HSE. Using HSE triggers a $\sim 10^{-2}$ transient and corrupts $\dot{M}$ measurements.

13.5 WKB amplitude (see B2)

$\delta v_\perp \propto (\rho v_A A)^{-1/2}$ (Poynting) or $(\rho v_A A)^{-1/4}$ (per-mode amplitude convention).

13.6 □ Verification

tests/test_mhd_wind_hse_stationary.cu — HSE atmosphere, 100 acoustic crossings, lock $\max |v_r|/c_s < 10^{-4}$.

14 B2. WKB wave action conservation for Alfvén waves

sympy script: scripts/b2_wave_action_wkb.py **verified:** dispersion $\omega^2 = v_A^2 k^2$; no-reflection limit under uniform $\rho_0, B_{r,0}$. **code checkpoints:** athena_mhd_kernels.cu::d_alfven_wave tests/test_mhd_wind_amplitude_scaling.cu.

14.1 Linearised transverse MHD

Around a static background $(\rho_0(r), B_{r,0}(r))$ with $v_{r,0} = 0$:

$$\partial_t v_\perp = \frac{B_{r,0}}{\rho_0} \partial_r B_\perp, \quad \partial_t B_\perp = B_{r,0} \partial_r v_\perp. \quad (\text{B2-linear})$$

WKB ansatz gives the Alfvén dispersion

$$\omega^2 = v_A^2 k^2, \quad v_A \equiv B_{r,0}/\sqrt{\rho_0}. \quad (\text{B2-disp})$$

Sympy verified that the determinant of the 2×2 linearised matrix equals $-\omega^2 + v_A^2 k^2$, giving the dispersion as the unique non-trivial mode.

14.2 Elsässer variables

$z_{\pm} \equiv v_{\perp} \mp B_{\perp}/\sqrt{\rho_0}$ decouple the outgoing and incoming Alfvén waves:

$$\partial_t z_{\pm} \pm v_A \partial_r z_{\pm} = S_{\text{refl}} z_{\mp}, \quad (\text{B2-Elsasser})$$

with a reflection source S_{refl} that depends on the background gradients. In a uniform $(\rho_0, B_{r,0})$, sympy verifies $S_{\text{refl}} = 0$ — no wave reflection in a uniform atmosphere (Alfvén waves are pure one-way in the uniform limit).

14.3 Wave action conservation

Outgoing z_+ in the absence of reflection conserves the wave action

$$\boxed{\rho_0 A |z_+|^2 / v_A = \text{const along characteristics.}} \quad (\text{B2-action})$$

For the amplitude-scaling law of a driver BC (relevant for Suzuki winds), **Poynting flux** $\rho_0 v_A A |\delta v_{\perp}|^2$ conservation gives

$$|\delta v_{\perp}| \propto (\rho_0 v_A A)^{-1/2}, \quad (\text{B2-Poynting})$$

while a **per-mode** amplitude normalisation gives the Jacques-1977 form

$$|\delta v_{\perp}| \propto (\rho_0 v_A A)^{-1/4}. \quad (\text{B2-Jacques})$$

The two differ because the former fixes the transmitted energy flux while the latter fixes the wave amplitude in Elsässer space. Suzuki wind codes use **the Poynting-flux convention** at the photospheric driver (amplitude $\langle \delta v_{\perp} \rangle \approx 1.25$ km/s in Shimizu+22) — make sure the kernel matches.

14.4 □ Verification

tests/test_mhd_wind_amplitude_scaling.cu — linearised Alfvén wave from a fixed BC driver. Measure $\delta v_{\perp}^{\text{rms}}(r)$ at $r = 5, 10, 20 R_*$; lock deviation from $(\rho v_A A)^{-1/2}$ at $< 5\%$.

15 B3. Parker critical point on a super-radial flux tube

sympy script: scripts/b3_parker_critical_point.py **verified:** Parker wind equation derivation; classical spherical limit $r_c = GM_*/(2c_s^2)$. **code checkpoints:** tests/test_mhd_parker_wind.cu.

15.1 Parker wind equation

Steady isothermal flow on a super-radial tube ($A(r) = r^2 f(r)$) with $p = c_s^2 \rho$ yields, after eliminating ρ via mass conservation:

$$\boxed{\left(v - \frac{c_s^2}{v}\right) \frac{dv}{dr} = c_s^2 \frac{d \ln A}{dr} - \frac{GM_*}{r^2}.} \quad (\text{B3-Parker})$$

Sympy derived from mass + momentum + EOS.

15.2 Critical (sonic) point

At $v = c_s$ the LHS vanishes, forcing

$$c_s^2 \left(\frac{2}{r_c} + \frac{d \ln f}{dr} \Big|_{r_c} \right) = \frac{GM_*}{r_c^2}. \quad (\text{B3-critical})$$

Spherical limit $f \equiv 1$: $r_c = GM_*/(2c_s^2)$ (classical Parker radius). Sympy verifies this by solving (B3-critical) analytically.

15.3 Asymptotic velocity

Far from r_c (spherical limit), $v(r) \sim c_s \sqrt{4 \ln(r/r_c) + \text{const}}$ — logarithmic growth, characteristic of the Parker isothermal wind.

15.4 □ Verification

tests/test_mhd_parker_wind.cu — isothermal, no magnetic driver, set $T = 2 \times 10^6$ K, $M_* = 1M_\odot$, measure Mach number crossing at $r = r_c$. Lock $|M(r_c) - 1| < 10^{-3}$.

16 B4. Well-balanced MHSE at the operator level

sympy script: scripts/b4_well_balanced_mhse.py **verified:** $F_{\text{wb}}(\mathbf{U}_{\text{hse}}) \equiv 0$ by construction; linear-wave Jacobian preserved, $\partial_{\mathbf{U}} F_{\text{wb}} = \partial_{\mathbf{U}} R$ at $\mathbf{U}_{\text{hse}}$ — perturbation dynamics unchanged. **code checkpoints:** athena_mhd_solver.cu::compute_residual_wb, athena_mhd_solver.cu::snapshot_hse_if_needed, tests/test_athena_mhd_wind_hse_stationary.cu.

16.1 The problem

Direct application of the discrete residual $R(\mathbf{U}) = -\partial_i \mathbf{F}_i + \mathbf{S}$ to a piecewise-constant MHSE atmosphere leaves a truncation residual of order $\mathcal{O}(\Delta r^2) \rho g$. For a Suzuki-style wind run this residual drives a $|\delta v_r|/c_s \sim 10^{-2}$ transient that corrupts the wind-mass-loss diagnostic for hundreds of crossing times.

16.2 Well-balanced residual (Bermúdez-Vázquez 1994 / Botta+04)

$$\boxed{F_{\text{wb}}(\mathbf{U}) \equiv R(\mathbf{U}) - R(\mathbf{U}_{\text{hse}})}.$$

By construction $F_{\text{wb}}(\mathbf{U}_{\text{hse}}) = 0$ (sympy-verified trivially). So an atmosphere at MHSE stays at MHSE to machine precision for arbitrarily long time, regardless of reconstruction order or Riemann solver.

16.3 Preservation of linear-wave dynamics

A small perturbation $\delta \mathbf{U}$ around the MHSE state:

$$F_{\text{wb}}(\mathbf{U}_{\text{hse}} + \delta \mathbf{U}) = \partial_{\mathbf{U}} R|_{\mathbf{U}_{\text{hse}}} \cdot \delta \mathbf{U} + \mathcal{O}(\delta \mathbf{U}^2).$$

Sympy-verified that the Jacobian of F_{wb} at $\mathbf{U}_{\text{hse}}$ **equals** the Jacobian of R at the same state. Thus linear MHD waves (Alfvén, magnetosonic) propagate through F_{wb} identically to R — no spurious damping or dispersion is introduced by the well-balancing subtraction.

16.4 What MHSE means in a super-radial flux tube

Reprise from §B1:

$$\partial_r p + \rho g + B_r^2 \partial_r (\ln A) = 0.$$

Note the **last term** is critical. A naïve WB that only cancels gravity (the hydrodynamic well-balancing) will leave an $\mathcal{O}(B_r^2)$ residual. The correction for the super-radial tube must include the area-divergence term.

16.5 Practical recipe

1. **Snapshot MHSE on the grid.** Integrate the ODE $\partial_r p_{\text{hse}} = -\rho_{\text{hse}} g - B_r^2 \partial_r (\ln A)$ using the given $(\rho_{\text{hse}}, B_r, A)$ profiles.
2. **Compute $R(\mathbf{U}_{\text{hse}})$ once** at startup, store per cell. (Assumes $\mathbf{U}_{\text{hse}}$ is time-independent — true for stationary background.)
3. **Evolve** $\partial_t \mathbf{U} = R(\mathbf{U}) - R(\mathbf{U}_{\text{hse}})$.
4. **Initial condition:** seed $\mathbf{U}^0 = \mathbf{U}_{\text{hse}}$. Initial RHS identically zero $\rightarrow$ no transient.

16.6 Failure modes observed elsewhere

- radial1d without this correction: Newton iteration looks converged ($|F| < 10^{-9}$) but the HSE slowly drifts at machine precision $\times$ resolution. Seen in pre-MS KH attempts: HSE-Newton is stable but cannot initiate KH because $F_v \equiv 0$ at MHSE.
- cart_ale2 with WB: integration-grade HSE stability for 10^4 crossings.

16.7 Snapshot invalidation

If $(\rho_{\text{hse}}, B_r, A)$ profiles change (e.g., user-driven parameter sweep), re-snapshot. The kernel can detect staleness via a hash of the profile arrays.

16.8 □ Verification checkpoints

- tests/test_athena_mhd_wind_hse_stationary.cu — MHSE atmosphere, 10^4 acoustic crossings, assert $\max |v_r|/c_s < 10^{-8}$ (with WB on) vs $\sim 10^{-2}$ (with WB off). The “off” case is a **positive control**: if turning off WB doesn’t produce a transient, the WB machinery is a no-op and should be audited.

17 C1. Ohmic (resistive) dissipation

sympy script: scripts/cl_ohmic_dissipation.py **verified:** $\nabla \times (\nabla \times B) = \nabla(\nabla \cdot B) - \nabla^2 B$ (all 3 components); $\nabla \times (\eta_O J) = \eta_O \nabla \times J + (\nabla \eta_O) \times J$. **code checkpoints:** athena_mhd_kernels.cu::d_ohmic_dissipation, tests/test_mhd_ohmic_diffusion.cu.

17.1 Ohm's law with finite conductivity

$$\mathbf{E} = -\mathbf{v} \times \mathbf{B} + \eta_O \mathbf{J}, \quad \mathbf{J} = \nabla \times \mathbf{B}. \quad (\text{C1-Ohm})$$

17.2 Induction equation

Constant η_O :

$$\partial_t \mathbf{B} = \nabla \times (\mathbf{v} \times \mathbf{B}) + \eta_O \nabla^2 \mathbf{B}. \quad (\text{C1-induction-const})$$

Variable $\eta_O(\mathbf{r})$:

$$\partial_t \mathbf{B} = \nabla \times (\mathbf{v} \times \mathbf{B}) + \eta_O \nabla^2 \mathbf{B} - (\nabla \eta_O) \times \mathbf{J}. \quad (\text{C1-induction-var})$$

Sympy verified via the vector identity $\nabla \times (\eta_O \mathbf{J}) = \eta_O \nabla \times \mathbf{J} + (\nabla \eta_O) \times \mathbf{J}$.

17.3 Joule heating

$$Q_{\text{Ohm}} = \eta_O |\mathbf{J}|^2 \geq 0. \quad (\text{C1-Q})$$

Manifestly non-negative; enters internal-energy source.

17.4 CFL for explicit diffusion

$$\Delta t \leq \frac{(\Delta x)^2}{2\eta_O} \text{ (1D)}, \frac{(\Delta x)^2}{4\eta_O} \text{ (2D)}, \frac{(\Delta x)^2}{6\eta_O} \text{ (3D)}. \quad (\text{C1-CFL})$$

At $r \approx 1000$ km in Matsuoka+24, $\eta_O \sim 10^{12}$ cm²/s and $\Delta x \sim 1$ km, so $\Delta t_{\text{Ohm}} \sim 10^{-4}$ s — much smaller than hydro CFL. In practice, Matsuoka+24 use **super-time-stepping** (Alexiades-Amiez-Gremaud 1996) to accelerate the explicit Ohmic update.

17.5 □ Verification

tests/test_mhd_ohmic_diffusion.cu — sinusoidal B_y perturbation, $\eta_O = \text{const}$. Lock $L^2(B_y)$ exponential decay rate matches analytical $e^{-\eta_O k^2 t}$ to $< 1\%$ at $N = 128$.

18 C2. Ambipolar diffusion (ion-neutral drift)

sympy script: scripts/c2_ambipolar_dissipation.py **verified:** $\mathbf{J}_\perp \cdot \hat{\mathbf{B}} = 0$; $\mathbf{J}_\parallel \times \mathbf{B} = 0$; $(\mathbf{J} \times \hat{\mathbf{B}}) \times \hat{\mathbf{B}} = -\mathbf{J}_\perp$; $(\mathbf{J} \times \mathbf{B}) \times \mathbf{B}/|\mathbf{B}|^2 = (\mathbf{J} \times \hat{\mathbf{B}}) \times \hat{\mathbf{B}}$; $Q_{\text{amb}} = \eta_A |\mathbf{J}_\perp|^2 \geq 0$. **code checkpoints:** athena_mhd_kernels.cu::d_ambipolar_flux, tests/test_mhd_ambipolar_bmin.cu.

18.1 Motivation

In a partially-ionised plasma, **neutrals** do not couple directly to $\mathbf{B}$; they feel the Lorentz force only via ion-neutral collisions. When the collision rate ν_{in} is slower than MHD timescales, ions drift relative to neutrals at

$$\mathbf{v}_{\text{drift}} = \mathbf{J} \times \mathbf{B} / (\rho_i \rho_n \nu_{in}),$$

yielding the non-ideal electric field

$$\mathbf{E}_{\text{amb}} = \eta_A (\mathbf{J} \times \hat{\mathbf{B}}) \times \hat{\mathbf{B}} = -\eta_A \mathbf{J}_{\perp}, \quad (\text{C2-Eamb})$$

with $\eta_A \equiv |\mathbf{B}|^2 / (\rho_i \rho_n \gamma_{in})$.

18.2 Parallel / perpendicular current selectivity

Critical property: only $\mathbf{J}_{\perp}$ is dissipated by ambipolar; the parallel current $\mathbf{J}_{\parallel}$ flows freely along $\hat{\mathbf{B}}$:

$$\mathbf{J}_{\parallel} \times \mathbf{B} = \mathbf{0}. \quad (\text{C2-selective})$$

18.3 Kernel-friendly form

For implementation, the non-unit-vector form is preferred:

$$\boxed{\mathbf{E}_{\text{amb}} = \eta_A \frac{(\mathbf{J} \times \mathbf{B}) \times \mathbf{B}}{|\mathbf{B}|^2}}, \quad (\text{C2-altform})$$

avoids dividing by $|\mathbf{B}|$ in the normalisation step — matters for low- $|\mathbf{B}|$ robustness (chromosphere bottoms).

18.4 Induction equation

$$\partial_t \mathbf{B} = \nabla \times (\mathbf{v} \times \mathbf{B}) + \nabla \times \left[\eta_A \frac{(\mathbf{J} \times \mathbf{B}) \times \mathbf{B}}{|\mathbf{B}|^2} \right]. \quad (\text{C2-induction})$$

Non-linear in $\mathbf{B}$ — key difference from Ohmic.

18.5 Ambipolar heating

$$\boxed{Q_{\text{amb}} = \eta_A |\mathbf{J}_{\perp}|^2 = \eta_A \frac{|\mathbf{J}|^2 |\mathbf{B}|^2 - (\mathbf{J} \cdot \mathbf{B})^2}{|\mathbf{B}|^2} \geq 0.} \quad (\text{C2-Q})$$

18.6 CFL

$$\Delta t \leq \frac{(\Delta x)^2}{2\eta_A |\mathbf{B}|^2 / \rho} \quad (\text{perdirection}). \quad (\text{C2-CFL})$$

Scales with $|\mathbf{B}|^2$ — tight CFL in strong-field regions.

18.7 □ Verification

tests/test_mhd_ambipolar_bmin.cu — B_x uniform, sinusoidal B_y with $\eta_A = \text{const}$. Lock decay rate against analytical; also verify that when $B_y \parallel B_x$ (i.e., $\mathbf{J} \parallel \mathbf{B}$), $Q_{\text{amb}} = 0$ and B_y does not decay. Catches any accidental scalar-diffusion substitution for the tensor form.

19 C3. Total-energy equation with non-ideal MHD

sympy script: `scripts/c3_resistive_energy.py` **verified:** ideal Poynting flux identity; $Q_{\text{Ohm}} = \eta_O |J|^2$; $Q_{\text{amb}} = \eta_A |J_\perp|^2$ (via Lagrange's identity). **code checkpoints:** `athena_mhd_kernels.cu::d_total_energy_flux`.

19.1 Total-energy equation

With both Ohmic + ambipolar, total energy is still conserved; only the Poynting flux gets a non-ideal addition:

$$\partial_t E + \nabla \cdot [(E + p^*)\mathbf{v} - \mathbf{B}(\mathbf{v} \cdot \mathbf{B}) + \mathbf{E}_{\text{ni}} \times \mathbf{B}] = 0, \quad (\text{C3-energy})$$

with $\mathbf{E}_{\text{ni}} = \eta_O \mathbf{J} + \eta_A (\mathbf{J} \times \mathbf{B}) \times \mathbf{B} / |\mathbf{B}|^2$.

19.2 Internal-energy source

The non-ideal dissipation becomes internal-energy heating:

$$\partial_t (\rho e) + \nabla \cdot (\rho e \mathbf{v}) = -p \nabla \cdot \mathbf{v} + Q_{\text{Ohm}} + Q_{\text{amb}}, \quad (\text{C3-internal})$$

$$Q_{\text{Ohm}} = \eta_O |\mathbf{J}|^2, \quad Q_{\text{amb}} = \eta_A |\mathbf{J}_\perp|^2 = \eta_A \frac{|\mathbf{J}|^2 |\mathbf{B}|^2 - (\mathbf{J} \cdot \mathbf{B})^2}{|\mathbf{B}|^2}.$$

Sympy verified the Lagrange identity $|\mathbf{J} \times \mathbf{B}|^2 = |\mathbf{J}|^2 |\mathbf{B}|^2 - (\mathbf{J} \cdot \mathbf{B})^2$ which maps between the two Q_{amb} forms.

19.3 Sign convention

$Q = -\mathbf{E}_{\text{ni}} \cdot \mathbf{J}$ is the rate at which EM energy is dissipated **into the fluid's internal energy** — positive for dissipative $\mathbf{E}_{\text{ni}}$. Both Ohmic and ambipolar give $Q > 0$.

19.4 Kernel structure

- **Total energy:** evolved via conservative flux (C3-energy). Non-ideal Poynting flux $\mathbf{E}_{\text{ni}} \times \mathbf{B}$ added to the hydro energy flux.
- **No direct Q kernel needed:** internal-energy source is implicitly accounted for by the flux divergence of the non-ideal Poynting term. Explicit computation of Q only needed for diagnostics.

19.5 □ Verification

`tests/test_mhd_ohmic_energy_conservation.cu` — uniform $\mathbf{B}_0$ with sinusoidal B_y perturbation + $\square_O = \square_{\text{const}}$. Measure total energy drift over 100 decay timescales. Lock $|\Delta E / E_0| < 10^{-8}$ — **any leak means the non-ideal Poynting flux term is missing from the energy flux**.

20 C4. Saha closure for η_O and η_A

sympy script: `scripts/c4_saha_ionization_closure.py` **verified:** Saha low-T limit $x_e \rightarrow 0$; high-T limit $x_e \rightarrow 1$. **code checkpoints:** `athena_mhd_kernels.cu::d_saha_eta, tests/test_mhd_saha_table.cu`.

20.1 Saha equation (pure hydrogen, LTE)

$$\boxed{\frac{x_e^2}{1 - x_e} = \frac{1}{n_H} \left(\frac{m_e k_B T}{2\pi \hbar^2} \right)^{3/2} e^{-\chi_H/k_B T}}, \quad (\text{C4-Saha})$$

with $\chi_H = 13.6$ eV and $x_e = n_e/(n_e + n_H)$. **Sympy verified** $x_e \rightarrow 0$ as $T \rightarrow 0$ and $x_e \rightarrow 1$ as $T \rightarrow \infty$.

20.2 Diffusivity closures (Draine 1983 / Choi+09)

Ohmic (electron-neutral collisions):

$$\eta_O \approx 234 \sqrt{T/10^4 \text{ K}} x_e^{-1} \text{ cm}^2/\text{s}. \quad (\text{C4-etaO})$$

Ambipolar (ion-neutral drift):

$$\eta_A = \frac{|\mathbf{B}|^2}{\rho_i \rho_n \gamma_{in}}, \quad \gamma_{in} \approx 3.5 \times 10^{13} \text{ cm}^3/\text{g/s}. \quad (\text{C4-etaA})$$

These are the forms used in Suzuki+25 (RGB winds) and Matsuoka+24 (solar chromosphere).

20.3 Weakly-ionised limit ($x_e \ll 1$)

$$\eta_O \sim \frac{234 \sqrt{T/10^4}}{x_e}, \quad \eta_A \sim \frac{|\mathbf{B}|^2}{x_e \rho^2 \gamma_{in}}.$$

Both diverge as $x_e \rightarrow 0$ — non-ideal effects are maximal at the chromosphere base where most hydrogen is neutral. This is why the RGB winds in Suzuki+25 get their $15\times \dot{M}$ suppression from ambipolar: it shuts down high-frequency Alfvén wave transmission at the chromospheric floor.

20.4 Magnetic Reynolds numbers

$$R_m^{\text{Ohm}} = \frac{Lv_A}{\eta_O}, \quad R_m^{\text{amb}} = \frac{Lv_A \rho}{\eta_A |\mathbf{B}|^2}.$$

Ideal MHD requires $R_m \gg 1$. Matsuoka+24 reports $R_m \sim 1 - 10$ at $r \approx 1000$ km in the solar chromosphere — non-ideal MHD kicks in below this altitude.

20.5 Kernel implementation

Pre-tabulate (η_O, η_A) as functions of $(\rho, T, |\mathbf{B}|)$ on a 3D table to avoid evaluating the Saha equation every timestep. Suzuki+25 uses a $128 \times 128 \times 64$ log-spaced table; cost is negligible once built.

20.6 □ Verification

tests/test_mhd_saha_table.cu — verify tabulated (η_O, η_A) match Draine 1983 Table 2 to $< 1\%$ at five reference (T, ρ) pairs: $(10^4, 10^{-9})$, $(6000, 10^{-7})$, $(4500, 10^{-8})$, $(3000, 10^{-11})$, $(10^5, 10^{-12})$.

21 C5. Sub-grid turbulent heating closure (Suzuki-Inutsuka 2005)

sympy script: scripts/c5_suzuki_turbulent_heating.py **verified:** positivity $\varepsilon_{\text{turb}} > 0$; pure-outward limit $|z^-| \rightarrow 0$ gives no cascade ($\varepsilon_{\text{turb}}^{\text{SY}} \rightarrow 0$); timescale bound $\Delta t \leq \lambda_{\text{cor}}/(2c_d|\delta v_\perp|)$. **code checkpoints:** athena_mhd_kernels.cu::d_turbulent_heating_source, tests/test_athena_mhd_turbulent_heating_positivity.cu.

21.1 Why this closure exists

The 1D super-radial flux-tube code (§B1) cannot resolve the 3D Alfvén wave turbulence that transfers outward-going wave energy to heat in the corona. Suzuki-Inutsuka 2005 (SI05) added a phenomenological sub-grid heating term that dissipates transverse wave energy at a prescribed cascade rate. This is the crucial ingredient without which the 1D wind models severely **under-estimate** coronal temperature.

21.2 Suzuki-Inutsuka 2005 form

$$\varepsilon_{\text{turb}}^{\text{SI}} = c_d \rho \frac{|\delta v_\perp|^3}{\lambda_{\text{cor}}}, \quad c_d \approx 0.1, \quad \lambda_{\text{cor}} \sim 10^7 \text{ cm.}$$

Dimensions: $[\rho][\delta v]^3/[\lambda] = \text{erg/cm}^3/\text{s}$ □.

21.3 Shoda-Yokoyama 2016 (Elsässer) form

$$\varepsilon_{\text{turb}}^{\text{SY}} = \frac{\rho}{2\lambda_{\text{cor}}} (|\mathbf{z}^+|^2|\mathbf{z}^-| + |\mathbf{z}^-|^2|\mathbf{z}^+|).$$

This form makes the **non-linear coupling** between counter-propagating modes explicit. Limit $|z^-| \rightarrow 0$ gives $\varepsilon_{\text{turb}}^{\text{SY}} \rightarrow 0$ (sympy-verified): no cascade without cross-polarisation — an essential physical consistency check.

21.4 Energy-equation coupling

$$\partial_t E + \nabla \cdot [(E + P^*)\mathbf{v} - \mathbf{B}(\mathbf{B} \cdot \mathbf{v})] = \varepsilon_{\text{turb}} \geq 0.$$

Positivity. Sympy-verified $\varepsilon_{\text{turb}} > 0$ for positive $\rho, \delta v_\perp, \lambda_{\text{cor}}, c_d$. The kernel must not inadvertently create a negative-definite heating source — any code review must check this.

21.5 Timescale bound (explicit-source CFL)

Per-step heating must not exceed the available wave KE:

$$\varepsilon_{\text{turb}} \cdot \Delta t \leq \frac{1}{2} \rho |\delta v_{\perp}|^2 \iff \Delta t \leq \frac{\lambda_{\text{cor}}}{2 c_d |\delta v_{\perp}|}.$$

In practice, for solar wind parameters ($|\delta v_{\perp}| \sim 30 \text{ km/s}$, $\lambda_{\text{cor}} \sim 10^8 \text{ cm}$), this bound is well above the hyperbolic CFL; no concern.

21.6 Shimizu+22 scaling

$$\lambda_{\text{cor}}(r) = \lambda_0 \sqrt{A(r)/A(R_*)}, \quad \lambda_0 \approx 10^7 \text{ cm}$$

(i.e., correlation length scales with the tube area because energy- containing eddies fill the tube cross-section). Replication of Shimizu+22 figures requires this scaling **and** $c_d = 0.1$ exactly — the paper’s calibration.

21.7 Extraction of $\delta v_{\perp}$

$\delta v_{\perp}$ is defined in the **Lagrangian / comoving frame**:

$$|\delta v_{\perp}|^2 \equiv \langle (v_y - \bar{v}_y)^2 + (v_z - \bar{v}_z)^2 \rangle,$$

with $\bar{v}$ a running time-average. In a 1D tube code with no transverse degree of freedom, this can instead be derived from the **Alfvén wave amplitude**

$$|\delta v_{\perp}|^2 = |z^+|^2 + |z^-|^2 - 2\mathbf{z}^+ \cdot \mathbf{z}^-$$

using the Elsässer variables tracked directly from §B2.

21.8 □ Verification checkpoints

- tests/test_athena_mhd_turbulent_heating_positivity.cu — random initial $(\rho, \delta v, \lambda)$ states, assert $\varepsilon_{\text{turb}} \geq 0$ across 100 samples; no NaN.
- tests/test_athena_mhd_suzuki_wind.cu — full replication of SI05 Fig. 2 mass-loss rate $\dot{M} \sim 2 \times 10^{-14} M_{\odot}/\text{yr}$ within $\pm 10\%$ (calibration tolerance per paper’s own sensitivity to c_d). Requires well-balanced MHSE (§B4).

22 C6. Spitzer-Härm anisotropic thermal conduction

sympy script: scripts/c6_spitzer_conduction.py **verified:** parallel projector $P_{\parallel} = \hat{\mathbf{b}}\hat{\mathbf{b}}^T$ idempotent with $\text{tr} P_{\parallel} = 1$; parallel / perpendicular decomposition of ∇T ; Kirchhoff potential identity $\mathbf{F}_c = -\nabla[\frac{2}{7}\kappa_0 T^{7/2}]$ in 1D; entropy production $\sigma_{\text{cond}} = \kappa_{\parallel}(\hat{\mathbf{b}} \cdot \nabla T)^2/T^2 \geq 0$; FTCS stability $\sigma \leq 1/2$. **code checkpoints:** athena_mhd_kernels.cu::d_spitzer_heat_flux, athena_mhd_solver.cu::compute_conduction_dt, tests/test_athena_mhd_conduction_isothermal.cu.

22.1 Why Spitzer conduction dominates the corona

In a fully-ionised, magnetised plasma the electron mean-free-path scales as $\lambda_e \propto T^2/n_e$, while the transit time between collisions $\tau_c \propto T^{3/2}/n_e$. Combining these with the electron thermal velocity $v_{th,e} \propto \sqrt{T}$ gives the Spitzer-Härm (1953) / Braginskii (1965) conductivity

$$\kappa_{\parallel}(T) = \kappa_0 T^{5/2}, \quad \kappa_0 \approx 10^{-6} \text{ erg cm}^{-1} \text{ s}^{-1} \text{ K}^{-7/2}.$$

Perpendicular to $\mathbf{B}$, each electron gyrates through $\Omega_c \tau_c \gg 1$ orbits between collisions, so the cross-field transport is suppressed by $(\Omega_c \tau_c)^{-2} \sim 10^{-18}$ in a coronal loop. To machine precision the effective conductivity is **rank-1**:

$$\mathbf{F}_c = -\kappa_{\parallel}(T) \hat{\mathbf{b}} (\hat{\mathbf{b}} \cdot \nabla T), \quad \hat{\mathbf{b}} = \mathbf{B}/|\mathbf{B}|. \quad (\text{C6-Fc})$$

22.2 Parallel / perpendicular decomposition

Sympy-verified the following on a general smooth $(T, \mathbf{B})$:

1. **Idempotent projector.** $P_{\parallel} \equiv \hat{\mathbf{b}}\hat{\mathbf{b}}^T$ satisfies $P_{\parallel}^2 = P_{\parallel}$ and $\text{tr } P_{\parallel} = 1$.
2. **Orthogonal decomposition.** $\nabla T = P_{\parallel} \nabla T + (I - P_{\parallel}) \nabla T$ with the second piece strictly perpendicular to $\hat{\mathbf{b}}$.
3. **Anisotropy bound.** For any ∇T the flux magnitude is capped: $|\mathbf{F}_c| = \kappa_{\parallel} |\hat{\mathbf{b}} \cdot \nabla T| \leq \kappa_{\parallel} |\nabla T|$.

22.3 Kirchhoff potential (1D closed form)

In the smooth isothermal-field (or B-aligned) limit the flux admits a **gradient-form** potential:

$$\mathbf{F}_c = -\nabla \left[\frac{2}{7} \kappa_0 T^{7/2} \right] \quad (1D / B\text{-aligned smooth flow}). \quad (\text{C6-Kirchhoff})$$

This is critical for implementation: the nonlinear flux can be written as $-\kappa_0 \partial_x (T^{7/2}) \cdot \frac{2}{7}$, which is a **linear** central-difference operator on the Kirchhoff-transformed variable $\Theta \equiv \frac{2}{7} \kappa_0 T^{7/2}$. Sympy verifies $\partial_x (\frac{2}{7} \kappa_0 T^{7/2}) = \kappa_0 T^{5/2} \partial_x T$ identically.

22.4 Low-density collisionless quench

At $\rho \lesssim 10^{-20} \text{ g cm}^{-3}$ the mean-free-path exceeds the scale height and the Spitzer formula over-predicts the flux by orders of magnitude (Gruzinov-Quataert 2004; Shoda+2018a). Suzuki 2203.15280 Eq. 12 / Shoda+2020 patch this with a phenomenological cutoff:

$$\mathbf{q}_{\text{cnd}} = -\min\left(1, \rho/\rho_{\text{cnd}}\right) (B_r/|\mathbf{B}|) \kappa_0 T^{5/2} \partial_r T, \quad \rho_{\text{cnd}} = 10^{-20} \text{ g cm}^{-3}. \quad (\text{C6-quench})$$

Do not drop the quench — unquenched Spitzer at coronal ρ drives Δt_{cond} to $\sim 10^{-4} \times$ the hyperbolic CFL (shown in Shimizu+22 Fig. 2a). The quench is a numerical device and must be on; flipping it off produces the correct physics (sharper thermal fronts) but the wall-clock cost is unmanageable.

22.5 Energy-equation coupling

The flux enters the total-energy equation conservatively:

$$\partial_t E + \nabla \cdot [(E + p^*)\mathbf{v} - \mathbf{B}(\mathbf{v} \cdot \mathbf{B}) + \mathbf{F}_c] = 0. \quad (\text{C6-energy})$$

Sign. Heat flows *down* the temperature gradient: $\mathbf{F}_c \cdot \nabla T = -\kappa_{\parallel} (\hat{\mathbf{b}} \cdot \nabla T)^2 \leq 0$ (sympy-verified). Entropy production is strictly non-negative:

$$\sigma_{\text{cond}} \equiv -\frac{\mathbf{F}_c \cdot \nabla T}{T^2} = \frac{\kappa_{\parallel}}{T^2} (\hat{\mathbf{b}} \cdot \nabla T)^2 \geq 0. \quad (\text{C6-entropy})$$

This is the 2nd-law certificate. Any kernel regression that flips a sign on $\mathbf{F}_c$ will show as $\sigma_{\text{cond}} < 0$ in the verification test.

22.6 Parabolic CFL (linearised FTCS)

Linearising around a background T_0 gives an effective thermal diffusivity

$$\chi_{\text{eff}} \equiv \frac{\kappa_0 T_0^{5/2}}{\rho c_v}.$$

Forward-Euler + central-space applied to $\partial_t T = \chi \partial_x^2 T$ has the amplification factor

$$g(\xi) = 1 - 4\sigma \sin^2(\xi/2), \quad \sigma \equiv \chi \Delta t / \Delta x^2. \quad (\text{C6-FTCS})$$

Sympy-verified: worst case $\xi = \pi$ gives $g = 1 - 4\sigma$; marginal stability at $\sigma = 1/2$ gives $g = -1$. For the Spitzer diffusivity the bound becomes

$$\Delta t_{\text{cond}} \leq \frac{1}{2} \min_{\text{cells}} \frac{\rho c_v \Delta x^2}{\kappa_0 T^{5/2}}. \quad (\text{C6-CFL})$$

At chromospheric parameters ($T \sim 10^5$ K, $\rho \sim 10^{-10}$ g cm $^{-3}$, $\Delta x \sim 50$ km) this gives $\Delta t_{\text{cond}} \sim 10^{-2}$ s, roughly a factor of 10^3 tighter than the hyperbolic CFL. At $T \sim 2 \times 10^6$ K the factor is $\sim 10^6$.

22.7 RKL2 super-time-stepping

Because the conductive CFL is so tight, all modern stellar-wind codes use **RKL2 super-time-stepping** (Meyer+2012, based on Alexiades-Amiez-Gremaud 1996). N Chebyshev sub-stages lift the explicit diffusion step up to

$$\Delta t_{\text{RKL2}} \approx \frac{N^2 + N}{4} \Delta t_{\text{cond}}. \quad (\text{C6-RKL2})$$

Per hydrodynamic step we pick $N = \lceil \sqrt{4\Delta t_{\text{hyp}} / \Delta t_{\text{cond}}} \rceil$ so that one RKL2 block spans the hydro step. The per-block cost is N conduction-flux evaluations; for $N = 20$ the conductive solver consumes $\sim 10\%$ of the wall-clock — negligible.

RKL2 is a **separate operator** applied after the hyperbolic step (operator splitting), **not** a modification of the VL2 integrator.

22.8 Implementation recipe (for athena_mhd future addition)

1. **Compute** T from primitive p/ρ and μ (partial-ionisation μ from §C4 Saha closure).
2. **Evaluate flux at faces:** $\mathbf{F}_c^{i\pm 1/2}$ using centred differences of T and the face-averaged $\hat{\mathbf{b}}$.
Key subtlety: $\hat{\mathbf{b}}$ is a unit vector — compute it by first averaging $\mathbf{B}$ to face centres, then normalising. Averaging $\hat{\mathbf{b}}$ after normalising at cell centres loses rotational symmetry.
3. **Apply quench** per (C6-quench).
4. **Add** $\nabla \cdot \mathbf{F}_c$ to the RHS of the energy equation.
5. **RKL2 sub-cycling** if $\Delta t_{\text{cond}} < 0.1 \Delta t_{\text{hyp}}$; otherwise integrate in-line with the hyperbolic step.

22.9 □ Verification checkpoints

- tests/test_athena_mhd_conduction_isothermal.cu — sinusoidal $T(x) = T_0(1 + A \cos kx)$ on a uniform $\mathbf{B}_0 = B_0 \hat{x}$ background, no flow. Lock $L^2(T - T_0)$ decay rate matches analytical $\exp(-\chi_{\text{eff}} k^2 t)$ to $< 1\%$ at $N = 128$.
- tests/test_athena_mhd_conduction_anisotropy.cu — same IC but with $\mathbf{B}_0 = B_0 \hat{y}$ (perpendicular to ∇T). Assert $L^2(T - T_0)$ stays constant over 100 collision times — cross-field quench to machine precision.
- tests/test_athena_mhd_conduction_kirchhoff.cu — in 1D with $T(x) \propto (1 + A \cos kx)$, compare evolution of $\frac{2}{7} \kappa_0 T^{7/2}$ versus direct nonlinear $\kappa_0 T^{5/2} \partial_x T$; lock relative difference $< 10^{-12}$.
- tests/test_athena_mhd_conduction_entropy.cu — random $(T, \mathbf{B}, \nabla T)$ states, 100 samples; assert $\sigma_{\text{cond}} \geq 0$ in every sample. Catches accidental sign flips.

Any failure on the anisotropy (perpendicular-quench) test is an immediate red flag: without rank-1 conductivity, 1D modelling of coronal loops (Aschwanden 2005 §4) overshoots T_{peak} by a factor of 2-3.

23 C7. Optically-thin radiative cooling

sympy script: scripts/c7_optically_thin_cooling.py **verified:** $\Lambda(T) > 0$ on piecewise power-law segments; τ_{cool} positive definite; Townsend 2009 closed-form $T(t) = [T_0^{1-\alpha} - C(1-\alpha)t]^{1/(1-\alpha)}$ numerically verified to satisfy $dT/dt = -CT^\alpha$ across 21 (α, T_0, C, t) samples (max err 0.0); $\alpha = 1$ degenerate exponential decay; logarithmic slope identity $d \ln \Lambda / d \ln T = \alpha$. **code checkpoints:** athena_mhd_kernels.cu::d_cooling_source, athena_mhd_solver.cu::apply_radiative_cooling, tests/test_athena_mhd_cooling_townsend.cu.

23.1 Regime of validity

Optically-thin cooling applies when the mean-free-path of emitted photons exceeds the local scale height, $\ell_{\text{photon}} \gtrsim H_\rho$. For a coronal plasma at $T \gtrsim 10^{5.2}$ K, $\rho \lesssim 10^{-13}$ g cm $^{-3}$ this is satisfied and the spectrum is dominated by bremsstrahlung + line emission (Sutherland-Dopita 1993, henceforth SD93).

Below $T \sim 1.5 \times 10^4$ K the chromosphere is **optically thick** to the dominant Lyman- α and Ca II lines; see §C8 for the blended thick-thin closure used in Suzuki+25 / Shimizu+22.

23.2 SD93 cooling rate

$$Q_R(T, \rho, Z) = n_e n_i \Lambda(T, Z) \geq 0, \quad n_e \approx n_i \approx \rho / (\mu_e m_u). \quad (\text{C7-QR})$$

$\Lambda(T, Z)$ is tabulated from the SD93 ionisation-equilibrium synthesis for $Z \in \{0, 10^{-3}, 10^{-2}, 10^{-1}, 1, 3\} Z_\odot$ over $\log T \in [4, 8.5]$. Piecewise power-law fit:

$$\Lambda(T) \approx \Lambda_k (T/T_k)^{\alpha_k}, \quad T \in [T_k, T_{k+1}]. \quad (\text{C7-piecewise})$$

Typical slopes (SD93 Table 6, solar Z):

Regime	$\log T$	α	Physics
Chromospheric tail	4.0-4.3	+3 to +5	HI, Ca II lines
Line-dominated	4.3-7.0	-0.5 to -1	Fe, O, Si
Bremsstrahlung	7.0-8.5	+1/2	free-free

23.3 Cooling timescale

$$\tau_{\text{cool}} = \frac{\varepsilon_{\text{th}}}{Q_R} = \frac{p}{(\gamma - 1) n_e n_i \Lambda(T)} = \frac{\mu_e^2 m_u^2 k_B T}{(\gamma - 1) \mu m_u \rho \Lambda(T)}. \quad (\text{C7-tau})$$

Sympy-verified $\tau_{\text{cool}} > 0$ for positive inputs.

Order-of-magnitude: at solar corona base ($T \sim 10^6 \text{ K}$, $\rho \sim 10^{-15} \text{ g cm}^{-3}$, $\Lambda \sim 10^{-22.7} \text{ erg cm}^3 \text{ s}^{-1}$) $\tau_{\text{cool}} \sim 10^5 \text{ s}$, comparable to a sound- crossing time over $\sim 1 R_\odot$. At the chromospheric transition region τ_{cool} drops to $\sim 10^0 \text{ s}$ — dramatically sub-CFL — motivating operator splitting with implicit / exactly-integrable sub-cycles (Townsend 2009).

23.4 Townsend 2009 closed-form integration

If $\Lambda(T) = \Lambda_0 (T/T_0)^\alpha$ on a power-law segment and n_e, ρ vary slowly (isochoric + slow- ρ splitting), the cooling ODE

$$\frac{dT}{dt} = -C T^\alpha, \quad C \equiv (\gamma - 1)(\mu m_u / k_B) n_e \Lambda_0 / T_0^\alpha > 0$$

admits the closed-form solution (Townsend 2009 Eq. 26):

$$T(t) = [T_0^{1-\alpha} - C(1-\alpha)t]^{1/(1-\alpha)} \quad (\alpha \neq 1). \quad (\text{C7-Townsend})$$

Numerically verified (21 samples across $\alpha \in \{-1, -\frac{1}{2}, 0, \frac{1}{2}, \frac{3}{2}, 2, 3\}$ and representative (T_0, C, t)) to satisfy $dT/dt + CT^\alpha = 0$ to machine precision.

Degenerate $\alpha = 1$ limit:

$$T(t) = T_0 \exp(-Ct). \quad (\text{C7-exp})$$

Kernel significance: on any time-step Δt_{hyp} that spans many τ_{cool} , (C7-Townsend) gives the *exact* sub-segment update — no sub-cycling required. This is the trick that makes Athena++ / PLUTO cooling kernels run at hyperbolic CFL instead of $0.1 \tau_{\text{cool}}$.

The full Townsend scheme handles crossing segment boundaries via a **temporal evolution function** $Y(T)$ that monotonically maps $T \mapsto$ integrated time; look-up + inversion stays closed-form.

23.5 Energy-equation coupling

$$\partial_t E + \nabla \cdot [\text{(hydro fluxes)}] = -Q_R(T, \rho, Z). \quad (\text{C7-energy})$$

Cooling is a **pure sink**: $Q_R \geq 0$ subtracts from internal energy. Sympy confirms $d \ln \Lambda / d \ln T = \alpha$ on each segment, so the Field 1965 isobaric stability criterion reads

$$\partial_T \Lambda|_p < 0 \iff \alpha < 2, \quad (\text{C7-field})$$

satisfied almost everywhere in $\log T \in [4.3, 7.0]$. This is the root of **thermal instability** in the ISM: the warm neutral / cold neutral bistability, and the Parker-like corona / chromosphere thermal segregation.

23.6 Explicit-source CFL (fallback, no Townsend)

When Townsend integration is disabled or the $\Lambda(T)$ table is not piecewise-power-law, sub-cycle:

$$\Delta t_{\text{rad}} \leq \beta_{\text{rad}} \tau_{\text{cool}}, \quad \beta_{\text{rad}} \approx 0.1. \quad (\text{C7-CFL})$$

$\beta_{\text{rad}} = 0.1$ is the Townsend 2009 calibrated safety margin; > 0.2 gives $\gtrsim 1\%$ errors in thermal-front propagation (SD93 shock tube benchmarks).

23.7 Metallicity scaling

SD93 $\Lambda(T, Z)$ tables decompose as

$$\Lambda(T, Z) = \Lambda_{\text{H,He}}(T) + (Z/Z_\odot) \Lambda_{\text{metals}}(T).$$

Two-parameter bilinear table lookup $\Lambda(\log T, \log Z)$ implemented as a 64×16 device texture; cost negligible once built (Suzuki+25 approach). For $Z < 10^{-4} Z_\odot$ (Pop III) the metal term is $\lesssim 10^{-3}$ of the total — atomic-H-only cooling tables (Anninos+1997) suffice instead.

23.8 Implementation recipe

1. **Tabulate** $\Lambda(\log T, \log Z)$ at build time or load from disk; store on GPU as 2D texture.
2. **Each step**: compute T from primitive (requires μ from §C4).
3. **Cooling sub-stage**: either (a) Townsend closed-form on the segment, or (b) explicit sub-cycle at $\beta_{\text{rad}} \tau_{\text{cool}}$.
4. **Apply** $\Delta E = -Q_R \cdot \Delta t$ after the hyperbolic step. No change to $\rho, \mathbf{v}, \mathbf{B}$ — the cooling is pure internal-energy relaxation.

5. **Safety floor:** clip $T \geq T_{\text{floor}}$ (e.g., $T_{\text{floor}} = 0.7T_{\text{eff}}$ in Suzuki+25) to prevent numerical runaway on under-resolved thermal fronts. The floor is a **known bias**, not a bug — document it in the driver log.

23.9 □ Verification checkpoints

- tests/test_athena_mhd_cooling_townsend.cu — uniform box, $\Lambda(T) = \Lambda_0(T/T_0)^{-1/2}$ (bremsstrahlung), fixed ρ . Assert $T(t)$ matches (C7-Townsend) at $t = \{0.1, 1, 10\} \tau_{\text{cool},0}$ to $< 10^{-10}$ relative error.
- tests/test_athena_mhd_cooling_segment_cross.cu — initial T_0 straddles two power-law segment boundaries; lock that the integrated $T(t)$ matches piecewise Townsend to $< 10^{-8}$.
- tests/test_athena_mhd_cooling_energy_floor.cu — pathological $T_0 < T_{\text{floor}}$, assert T sticks at floor and no NaN.
- tests/test_athena_mhd_cooling_field_instability.cu — seed an isobaric T perturbation in a Field-unstable regime ($\alpha = -1$); lock linear growth rate to analytic $\sigma = |\alpha - 2|/(2\tau_{\text{cool},0})$ within 5%.

The Field-instability test is the second-law certificate: any regression that drops the Q_R sign will mistake cooling for heating and wreck this test first.

24 C8. Chromospheric (optically-thick) cooling and blended closure

sympy script: scripts/c8_chromospheric_cooling.py **verified:** blend weight $\xi_{\text{rad}} \in [0, 1)$ with anchor at $p = p_{\text{chr}}$; Newton cooling has closed-form exponential solution $T(t) = T_{\text{ref}} + (T_0 - T_{\text{ref}}) \exp(-t/\tau_{\text{thck}})$; GN05 scaling $d \ln \tau_{\text{thck}} / d \ln \rho = -1/2$; Anderson-Athay cooling is monotone in Z , saturated above ρ_{cr} ; convex-combination partials $\partial_{Q_{\text{thck}}} Q_R = \xi$, $\partial_{Q_{\text{thin}}} Q_R = 1 - \xi$. **code checkpoints:** athena_mhd_kernels.cu::d_chromo_newton_cooling, athena_mhd_kernels.cu::d_anderson_athay_cooling, athena_mhd_solver.cu::apply_blended_cooling, tests/test_athena_mhd_chromo_relax.cu.

24.1 Why the chromosphere needs its own closure

At $T \lesssim 1.5 \times 10^4$ K the dominant radiative loss channels (Lyman- α , Ca II H+K, Mg II h+k, Balmer continuum) are **saturated** — the emergent intensity equals the Planck function at the optical-surface temperature, so cooling becomes independent of the *local* $n_e n_i$ and instead relaxes toward the photospheric T_{eff} . The Sutherland-Dopita 1993 optically-thin rate (§C7) overestimates cooling by $10^4 \times$ in this regime (Vernazza-Avrett-Loeser 1981 VAL3 atmosphere). Suzuki-group codes patch this with a two-component closure.

24.2 Shimizu+22 blending formula

$$Q_R(T, \rho, p) = \xi_{\text{rad}}(p) Q_R^{\text{thck}}(T, \rho) + (1 - \xi_{\text{rad}}(p)) Q_R^{\text{thin}}(T, \rho, Z). \quad (\text{C8-blend})$$

with pressure-switch blending

$$\xi_{\text{rad}}(p) = \max(0, 1 - p_{\text{chr}}/p), \quad p_{\text{chr}} \approx 0.1 p_{\odot}. \quad (\text{C8-xi})$$

Physical interpretation. In the dense chromosphere $p \gg p_{\text{chr}}$ gives $\xi \rightarrow 1$ (pure Newton cooling toward the photospheric temperature). In the tenuous corona $p \lesssim p_{\text{chr}}$ gives $\xi \rightarrow 0$ (pure optically-thin SD93). The transition region at $T \sim 15000$ K is where $p \sim p_{\text{chr}}$ and the closure smoothly blends both channels.

Sympy-verified $\xi_{\text{rad}} \in [0, 1)$ and anchors at zero at the cutoff $p = p_{\text{chr}}$.

24.3 Gudiksen-Nordlund 2005 Newton cooling

$$Q_R^{\text{thck}} = \frac{e_{\text{int}} - e_{\text{int}}^{\text{ref}}(r)}{\tau_{\text{thck}}(\rho)}, \quad \tau_{\text{thck}}(\rho) = 0.1 (\rho/\bar{\rho})^{-1/2} \text{ s}, \quad (\text{C8-GN05})$$

with $\bar{\rho} = 1.87 \times 10^{-7} \text{ g cm}^{-3}$ (Shimizu+22 calibration) and $T^{\text{ref}}(r) = T_{\odot}$ (or T_{eff} in generic stars).

Exponential relaxation. For fixed ρ , the ODE $dT/dt = -(T - T_{\text{ref}})/\tau_{\text{thck}}$ has the closed-form solution

$$T(t) = T_{\text{ref}} + (T_0 - T_{\text{ref}}) \exp(-t/\tau_{\text{thck}}). \quad (\text{C8-relax})$$

Sympy-verified by direct substitution. The GN05 scaling $\tau_{\text{thck}} \propto \rho^{-1/2}$ (sympy: log-slope = $-1/2$) lengthens relaxation in tenuous layers — crucial to keep chromospheric shock-train structures from being over-damped.

24.4 Anderson-Athay 1989 empirical chromospheric rate

An independent branch used in Suzuki-Ohnaka-Yasuda 2025 (eq. 17):

$$Q_R^{\text{AA}}(\rho, Z) = 4.5 \times 10^9 (0.2 + 0.8 Z/Z_{\odot}) \min(1, \rho/\rho_{\text{cr}}) \text{ erg cm}^{-3} \text{ s}^{-1}, \quad \rho_{\text{cr}} = 10^{-16} \text{ g cm}^{-3}. \quad (\text{C8-AA})$$

Properties (sympy-verified): - Strictly non-negative for $\rho, Z \geq 0$. - Monotone increasing in $Z/Z_{\odot}$: $\partial_Z Q_{\text{AA}} = 3.6 \times 10^9 \cdot \min(1, \rho/\rho_{\text{cr}}) \geq 0$. - Saturates at $\rho \geq \rho_{\text{cr}}$: above the critical density, the chromosphere becomes fully optically thick and the cooling rate is density-independent (a photospheric-photon property).

The Anderson-Athay form is **not** exchangeable with the GN05 Newton form — the former is a *rate* (per volume), the latter is a *relaxation toward a reference state*. The Suzuki+25 code selects AA for RGB-wind atmospheres, Shimizu+22 selects GN05 for solar wind.

24.5 Two-piece thin cooling piece

$$Q_R^{\text{thin}}(T, \rho, Z) = \begin{cases} Q_R^{\text{SD93}}(T, Z) n_e n_i, & T > 1.2 \times 10^4 \text{ K} \\ Q_R^{\text{AA}}(\rho, Z), & T \leq 1.2 \times 10^4 \text{ K} \\ 0, & T \leq T_{\text{cut}} = 0.7 T_{\text{eff}} \end{cases} \quad (\text{C8-twopiece})$$

T_{cut} is a **numerical-stability floor**, not a physical cutoff. Without it the chromospheric rate diverges at the photospheric boundary and ejects the density to $\rho \rightarrow 0$. The $0.7 T_{\text{eff}}$ threshold is Suzuki+25 calibration (Sec 2.5); at $T_{\text{cut}} = 3010$ K for α Boo.

24.6 Convex combination preserves positivity

For $\xi_{\text{rad}} \in [0, 1]$ and $Q_R^{\text{thick}}, Q_R^{\text{thin}} \geq 0$:

$$Q_R = \xi Q_R^{\text{thick}} + (1 - \xi) Q_R^{\text{thin}} \geq 0. \quad (\text{C8-positivity})$$

Sympy partial derivatives confirm $\partial_{Q_{\text{thick}}} Q_R = \xi$ and $\partial_{Q_{\text{thin}}} Q_R = 1 - \xi$ — the blend is *linear in each component*, so the combined cooling inherits positivity from its parts.

24.7 Implicit update (backward-Euler Newton cooling)

Because τ_{thick} can drop below the hyperbolic CFL in dense regions, the GN05 branch is usually integrated **implicitly**. For fixed ρ the single-step Backward-Euler reduces to

$$T^{n+1} = \frac{T^n + \nu T_{\text{ref}}}{1 + \nu}, \quad \nu \equiv \Delta t / \tau_{\text{thick}}, \quad (\text{C8-BE})$$

which is **unconditionally stable** and exactly monotone toward T_{ref} . Kernel cost: one division per cell. No GMRES / Newton-Krylov required because Q_R^{thick} is linear in T .

The thin-cooling branch uses Townsend 2009 (§C7 C7-Townsend); the combined step applies BE for the thick piece and Townsend for the thin piece in operator-split fashion.

24.8 Smooth blending (optional)

The piecewise ξ_{rad} of (C8-xi) is C^0 at $p = p_{\text{chr}}$. For applications sensitive to the smoothness of Q_R (e.g., linear g-mode propagation through the transition region), a C^∞ alternative is

$$\xi_{\text{rad}}^{\text{smooth}}(p) = \frac{1}{2} [1 + \tanh((p - p_{\text{chr}}) / \Delta p)]. \quad (\text{C8-smooth})$$

$\Delta p \sim p_{\text{chr}} / 10$ keeps the transition width small. Shimizu+22 uses the piecewise form; Suzuki+25 uses neither (pure AA below the $T = 1.2 \times 10^4$ threshold, no blending).

24.9 Explicit vs implicit CFL comparison

Branch	Explicit CFL	Implicit CFL
GN05 Newton (thick)	$\Delta t \leq 0.3 \tau_{\text{thick}}$	unconditional
SD93 (thin)	$\Delta t \leq 0.1 \tau_{\text{cool}}$	Townsend exact
AA chromosphere	depends on ρ ; trivially sub-CFL	direct multiply

Recommendation: in the kernel, use BE for GN05 and Townsend for SD93 in an operator-split cooling pass applied after the hyperbolic step. Neither branch touches $\rho, \mathbf{v}, \mathbf{B}$ — only internal energy.

24.10 □ Verification checkpoints

- tests/test_athena_mhd_chromo_relax.cu — 1D isobaric atmosphere seeded at $T_0 = 6000$ K above $T_{\text{ref}} = 5770$ K with $\tau_{\text{thck}} = 0.1$ s; lock that $T(t)$ matches (C8-relax) at $t = \{0.1, 1, 10\} \tau_{\text{thck}}$ to $< 10^{-10}$ relative error.
- tests/test_athena_mhd_chromo_blend.cu — 1D atmosphere spanning $T \in [3000, 2 \times 10^6]$ K; assert $\xi_{\text{rad}}(p)$ profile matches (C8-xi) cell-by-cell; assert smoothness / monotonicity.
- tests/test_athena_mhd_chromo_positivity.cu — 100 random (T, ρ, p, Z) samples, assert $Q_R \geq 0$ and no NaN.
- tests/test_athena_mhd_chromo_floor.cu — atmosphere cooled past $T_{\text{cut}} = 0.7 T_{\text{eff}}$; assert T stays at floor and $Q_R^{\text{thin}} = 0$ there.

The positivity test is the most important — it catches any sign-flip regression in the AA or SD93 branches, which would let the code “heat via radiation” and crash.

25 D1. Ideal MHD in cylindrical (R, ϕ, z) coordinates

sympy script: scripts/d1_cylindrical_mhd.py **verified:** displays cylindrical divergence, curl, and axisymmetric reductions. No new nontrivial identities beyond Part A; this section is the reference for D2 and D3. **code checkpoints:** athena_mhd_cylindrical_solver.cu (future).

25.1 Cylindrical operators

$$\nabla \cdot \mathbf{v} = \frac{1}{R} \partial_R (R v_R) + \frac{1}{R} \partial_\phi v_\phi + \partial_z v_z.$$

$$(\nabla \times \mathbf{B})_R = \frac{1}{R} \partial_\phi B_z - \partial_z B_\phi, \quad (\nabla \times \mathbf{B})_\phi = \partial_z B_R - \partial_R B_z, \quad (\nabla \times \mathbf{B})_z = \frac{1}{R} [\partial_R (R B_\phi) - \partial_\phi B_R].$$

25.2 Radial momentum (axisymmetric)

$$\partial_t (\rho v_R) + \frac{1}{R} \partial_R (R \rho v_R^2) + \partial_z (\rho v_R v_z) = \boxed{\frac{\rho v_\phi^2}{R}} - \partial_R p - \rho \partial_R \Phi_{\text{grav}}. \quad (\text{D1-mom-R})$$

The $\rho v_\phi^2 / R$ term is the **Christoffel-symbol / centrifugal source**. It is what keeps a rotating disk in equilibrium against gravity. In a kernel, it is the most common source of subtle bugs — easy to drop when porting Cartesian kernels.

25.3 Axisymmetric radial-flux conservation

$\partial_\phi = \partial_z = 0$ with $\mathbf{B} = B_R \hat{R}$ forces $\nabla \cdot \mathbf{B} = (1/R) \partial_R (R B_R) = 0$, so $R \cdot B_R(R) = \text{const}$ — the cylindrical analog of the spherical $r^2 B_r$ conservation.

25.4 □ Verification

tests/test_mhd_cyl_centrifugal.cu — rotating equilibrium disk with $v_\phi = \sqrt{GM/R}$. Lock $\max |v_R| / c_s < 10^{-4}$ over 10 rotations. Failure means the centrifugal source term is missing or mis-signed.

26 D2. Shearing-sheet approximation and shearing-periodic BC

sympy script: `scripts/d2_shearing_sheet_bc.py` **verified:** shearing-sheet background is steady state (Coriolis + tidal cancel on v_ϕ^{bg}); shear wrap-around time $t_{\text{shear}} = L_y/(q\Omega_0 L_x)$; effective potential is anti-confining. **code checkpoints:** `athena_mhd_shearingbox_bc.cu`, `tests/test_mhd_shearingbox_static.cu`.

26.1 Hill expansion

Local Cartesian patch centred at (R_0, Ω_0) , co-rotating with the disk. Linearised rotational velocity:

$$v_\phi^{\text{bg}}(x) = -q\Omega_0 x, \quad q \equiv -\frac{d \ln \Omega}{d \ln R}. \quad (\text{D2-shear})$$

$q = 3/2$ for Keplerian disks, $q = 2$ for rigid rotation.

26.2 Frame-rotation sources

Coriolis:

$$\mathbf{F}_{\text{Cor}} = -2\Omega_0 \hat{z} \times \mathbf{u} = 2\Omega_0(u_y \hat{x} - u_x \hat{y}).$$

Tidal (Hill):

$$\mathbf{F}_{\text{tidal}} = 2q\Omega_0^2 x \hat{x} = -\nabla \Phi_{\text{eff}}, \quad \Phi_{\text{eff}} = -q\Omega_0^2 x^2.$$

Sympy verified that on the background $\mathbf{u}_{\text{bg}} = (0, -q\Omega_0 x, 0)$, $F_{\text{Cor},x} + F_{\text{tidal},x} = 0$. The shearing-sheet background is an exact steady state of the force-balance.

26.3 Shearing-periodic BC

$$\text{field}(x = L_x, y, z, t) = \text{field}(x = 0, y - q\Omega_0 L_x t, z, t). \quad (\text{D2-BC})$$

The y -shift $\Delta y(t) = q\Omega_0 L_x t$ grows linearly in time. After $t_{\text{shear}} = L_y/(q\Omega_0 L_x)$ the shift wraps to L_y and the BC becomes pure periodic again.

Under periodic- y convention, the BC is always well-defined: take $\Delta y \bmod L_y$. **Sympy verified** t_{shear} .

26.4 Effective potential is anti-confining

$$\frac{d^2 \Phi_{\text{eff}}}{dx^2} = -2q\Omega_0^2 < 0,$$

so the tidal potential alone is unstable — **the Coriolis force is what stabilises the shearing sheet**. Sympy-verified; documented to flag that dropping the Coriolis kernel would immediately disperse the disk.

26.5 □ Verification

tests/test_mhd_shearingbox_static.cu — initial condition $\mathbf{u} = (0, -q\Omega_0 x, 0)$, purely hydro, no perturbation. Lock max perturbation energy $< 10^{-6} \times E_0$ over $10/\Omega_0$. Any drift means Coriolis or tidal coupling is mis-implemented.

27 D3. MRI stress decomposition $\alpha_{\text{SS}} = \alpha_R + \alpha_M$

sympy script: scripts/d3_mri_stress_decomposition.py **verified:** Maxwell stress antisymmetric under $B_R \rightarrow -B_R$, invariant under $(B_R, B_\phi) \rightarrow (-B_R, -B_\phi)$; Reynolds stress antisymmetric under $v_R \rightarrow -v_R$; MRI dispersion vanishes at Balbus-Hawley extremum ($k_*^2 v_A^2 = 15/16 \Omega^2$, $\gamma_{\text{max}} = 3/4 \Omega$). **code checkpoints:** tests/test_mhd_mri_growth_rate.cu, tests/test_mhd_mri_stress_decomp.cu.

27.1 Shakura-Sunyaev stress

$$T_{R\phi} = \underbrace{\rho v_R \delta v_\phi}_{\text{Reynolds}} + \underbrace{(-B_R B_\phi)}_{\text{Maxwell}}. \quad (\text{D3-stress})$$

$$\alpha_{\text{SS}} = \frac{\langle T_{R\phi} \rangle}{\langle p \rangle} = \alpha_R + \alpha_M.$$

27.2 Suzuki 2023 sign-quadrant decomposition

Maxwell stress is: - **antisymmetric** under single flip $B_R \rightarrow -B_R$: $M_{R\phi} \rightarrow -M_{R\phi}$. - **invariant** under simultaneous flip $(B_R, B_\phi) \rightarrow (-B_R, -B_\phi)$.

This pair of symmetries is what makes the Suzuki 2305.12112 “triangle diagnostic” meaningful: the four sign quadrants $(\pm B_R, \pm B_\phi)$ carry physically distinct stress contributions. The cylindrical vs Cartesian sign-flip result in that paper (headline $[\phi \Rightarrow_R \phi]$ arrow +6.42 vs −1.05) is a direct consequence of this symmetry structure.

27.3 Balbus-Hawley MRI dispersion

For vertical $\mathbf{B}_0 = B_0 \hat{z}$ and Keplerian $q = 3/2$ ($\kappa^2 = \Omega^2$):

$$\omega^4 - \omega^2(\kappa^2 + 2k^2 v_A^2) + k^2 v_A^2 (k^2 v_A^2 + \kappa^2 - 4\Omega^2) = 0. \quad (\text{D3-BH-disp})$$

Instability requires $k^2 v_A^2 < 3\Omega^2$. The most-unstable mode:

$$\boxed{k_*^2 v_A^2 = \frac{15}{16} \Omega^2, \quad \gamma_{\text{max}} = \frac{3}{4} \Omega.} \quad (\text{D3-MRI-max})$$

Sympy verified by direct substitution into the dispersion polynomial. This gives the MRI timescale $\sim 4/(3\Omega) \approx 1/3$ of an orbital period.

27.4 Parseval for Maxwell

$$\frac{1}{V} \int B_R(\mathbf{x}) B_\phi(\mathbf{x}) d^3x = \sum_{\mathbf{k}} \text{Re}[\hat{B}_R(\mathbf{k}) \hat{B}_\phi^*(\mathbf{k})]$$

Allows FFT-based computation of α_M at arbitrary resolution.

27.5 $\square$ Verification

tests/test_mhd_mri_growth_rate.cu — seed a linear B_y perturbation at k_* with amplitude $10^{-6} B_0$; lock measured growth rate matches $(3/4)\Omega$ to $< 1\%$ over $5/\Omega$.

tests/test_mhd_mri_stress_decomp.cu — in nonlinear saturated state, lock $\alpha_M/\alpha_R \sim 3 - 5$ (Stone+96, Suzuki+23 Cartesian baseline). Suzuki+23 reports $\alpha_M \approx 0.072$, $\alpha_R \approx 0.016$ for Cartesian Keplerian baseline.

28 E1. Stochastic broadband photospheric driver

sympy script: `scripts/el_stochastic_driver.py` **verified:** $\int_{\omega_{\min}}^{\omega_{\max}} (A^2/\omega) d\omega = A^2 \ln(\omega_{\max}/\omega_{\min})$; target-variance normalisation $A^2 = \langle \delta v^2 \rangle / \ln(\omega_{\max}/\omega_{\min})$; single-sinusoid time-variance $\langle \sin^2 \rangle_t = 1/2$; cross-frequency terms vanish in time average (Parseval); linearised Elsässer $\partial_t z^\pm \pm v_A \partial_r z^\pm = 0$ (pure one-way advection around uniform background); zero-gradient BC on z^- forces incoming amplitude to zero (absorbing / no reflection in WKB limit). **code checkpoints:** `athena_mhd_kernels.cu::d_photospheric_driver`, `athena_mhd_kernels.cu::d_absorbing_bc_zminus`, `tests/test_athena_mhd_driver_spectrum.cu`.

28.1 Why a broadband stochastic driver

Suzuki-group solar / red-giant wind simulations require continuous wave injection at the inner boundary to replace the granulation-driven photospheric convection. The driver must

1. **Carry the correct Poynting flux** so that the corona is heated to observed $T \sim 10^6$ K.
2. **Span a broadband spectrum** because phenomenological sub-grid turbulence (§C5) needs a realistic range of interacting frequencies to cascade.
3. **Be stochastic in phase** so that coherent resonances with the global domain modes do not build up.
4. **Pair with an absorbing BC for the incoming Alfvén wave** so that coronal material reflecting back down does not artificially amplify the driver.

28.2 Driver form (Shimizu+22 eq. 38 / 42; Suzuki+25 Sec 2.8)

The transverse Elsässer variable outgoing from the photosphere is driven as

$$z_{\perp, \odot}^+(t) = A_\perp \sum_{N=0}^{N_{\max}} \frac{\sin(2\pi f_N t + \varphi_N)}{\sqrt{f_N}}, \quad \varphi_N \sim U[0, 2\pi). \quad (\text{E1-driver-transverse})$$

The longitudinal radial velocity is driven independently:

$$\delta v_{\parallel, \odot}(t) = A_{\parallel} \sum_{N=0}^{N_{\max}} \frac{\sin(2\pi f_N^{\parallel} t + \varphi_N^{\parallel})}{\sqrt{f_N^{\parallel}}}. \quad (\text{E1-driver-long})$$

Log-spaced sampling mimics the continuous ω^{-1} spectrum:

$$f_N = f_{\min} (f_{\max}/f_{\min})^{N/N_{\max}}, \quad f_{\max} = 100 f_{\min}. \quad (\text{E1-logspace})$$

Suzuki-group calibration: - Transverse (Alfvén): $f_{\min} \approx 10^{-3}$ Hz, $f_{\max} \approx 10^{-2}$ Hz (16.7 min - 100 s); - Longitudinal (p-mode): $f_{\min}^{\parallel} \approx 3.33 \times 10^{-3}$ Hz, $f_{\max}^{\parallel} \approx 10^{-2}$ Hz (5 min - 100 s).

28.3 Power spectrum and normalisation

The $1/\sqrt{f_N}$ amplitude weighting reproduces a **flat-power-per- log-octave** spectrum: $P(\omega) \propto 1/\omega$. Sympy-verified:

$$\int_{\omega_{\min}}^{\omega_{\max}} \frac{A^2}{\omega} d\omega = A^2 \ln(\omega_{\max}/\omega_{\min}). \quad (\text{E1-spectrum})$$

Normalising to the observed target rms fluctuation (Suzuki+25 solar calibration: $\langle \delta v_{\perp} \rangle_{\odot} = 1.25$ km/s; RGB α Boo: 2.50 km/s via $\delta v \propto (T_{\text{eff}}^4/\rho)^{1/3}$):

$$A^2 = \frac{\langle \delta v^2 \rangle}{\ln(\omega_{\max}/\omega_{\min})}, \quad \int P(\omega) d\omega = \langle \delta v^2 \rangle. \quad (\text{E1-norm})$$

28.4 Parseval variance identity

For a sum of sinusoids with distinct frequencies and independent phases:

$$\langle [\sum_N A_N \sin(\omega_N t + \varphi_N)]^2 \rangle_t = \frac{1}{2} \sum_N A_N^2. \quad (\text{E1-parseval})$$

Sympy-verified symbolically on the time integral over a common period: single-sinusoid variance is $A^2/2$, cross-terms integrate to zero. Combining with $A_N = A/\sqrt{f_N}$ gives

$$\sum_N A_N^2 = A^2 \sum_N 1/f_N \approx A^2 \ln(f_{\max}/f_{\min})$$

(discrete approximation of the continuous integral).

The factor of $1/2$ means **the driver must be pre-multiplied by $\sqrt{2}$** to hit $\langle \delta v^2 \rangle$ — this is the source of the explicit $\sqrt{2}$ factor in the Shimizu+22 driver (their footnote to Eq. 41).

28.5 Random phases — why essential

If φ_N are deterministic, the driver is periodic with period $T_{\min} = 2\pi / \gcd(\omega_N)$ (for rational ω_N ratios) or quasi-periodic. Modes commensurate with the domain resonate — in the solar wind problem this shows up as spurious peaks in the coronal temperature spectrum.

Phase regeneration frequency. Draw new φ_N each time one simulation-correlation time elapses; practical choice $\Delta t_\varphi \sim 10/f_{\min}$. Alternative: draw once per run with a fixed random seed, document the seed in run metadata (Suzuki+25 approach — makes individual runs reproducible).

28.6 Absorbing BC for incoming Alfvén

The coronal run generates downward-propagating Alfvén modes that must not pile up at the photosphere. The physically-correct boundary condition is the **free-outgoing** form in Elsässer variables:

$$\boxed{\partial_r z_\perp^-|_{r=R_*} = 0.} \quad (\text{E1-BC})$$

For a plane wave $z^-(r, t) = Z_0 e^{i(kr - \omega t)}$, this forces $ik Z_0 = 0$ — i.e., $Z_0 = 0$ (sympy-verified via solve). In WKB: **zero reflection**, all incoming wave energy is absorbed.

Kernel form. Apply zero-gradient copy on the incoming Elsässer variable at the inner ghost cells:

$$z_\perp^-(r = R_* - \text{ghost}) \leftarrow z_\perp^-(r = R_* + 1 \text{ cell}).$$

Outgoing $z_\perp^+$ is clamped to the driver value (E1-driver-transverse). Both together decouple the driver from the interior reflection.

28.7 Elsässer propagation check (background dynamics)

Around a uniform static background $(\rho_0, B_{r,0})$, the linearised transverse equations reduce to **pure one-way advection** of the Elsässer variables:

$$\partial_t z^\pm \pm v_A \partial_r z^\pm = 0, \quad v_A = B_{r,0} / \sqrt{\rho_0}. \quad (\text{E1-advect})$$

Sympy-verified directly by substituting $z^\pm = v_\perp \mp B_\perp / \sqrt{\rho_0}$ into the linearised induction + momentum equations.

Consequence: the driver at $r = R_*$ couples *only* to z^+ ; corrections from finite background gradients (refraction, reflection, mode conversion) appear at $\mathcal{O}(\Delta r / \lambda_A)$ and are **part of the physical solution**, not BC artifacts. This is the content of Parker (1965) and Hollweg (1981) WKB treatments.

28.8 Driver implementation recipe

```
// One-time initialisation at kernel start-up
srand(seed);
for (N = 0; N < N_max; ++N) {
    f[N] = f_min * pow(f_max/f_min, double(N)/N_max); // log-spaced
    phi[N] = 2*M_PI * rand01();                       // uniform U[0,2π)
```

```

}
double A_perp = sqrt(2) * <dv_rms> / sqrt(log(f_max/f_min));

// Per time-step at inner ghost
double zplus = 0.0;
for (N = 0; N < N_max; ++N) {
    zplus += A_perp * sin(2*M_PI*f[N]*t + phi[N]) / sqrt(f[N]);
}
// Absorbing BC: copy interior value onto incoming Elsässer ghost
zminus_ghost = zminus_interior_lcell;
// Reconstruct v_perp, B_perp from (z+, z-)
v_perp = 0.5*(zplus + zminus_ghost);
B_perp = 0.5*sqrt(rho_0) * (zminus_ghost - zplus);

```

Key points: - **Compute $A_\perp$ once**, not per-step; the log-factor is fixed. - **$\sqrt{2}$ is important**; Parseval demands it. - **Absorbing BC copies one cell inward** (not two, not zeroth-order extrapolation). Two-cell copy is fine but a zeroth-order extrapolation will produce a small-amplitude reflection. - **Low-frequency end $f_{\min}^{-1}$ sets the run length floor**: one $f_{\min}^{-1}$ period must fit within the simulation, else the driver is aliased.

28.9 Special case: acoustic-only (no transverse driver)

Shimizu+22 Appendix A runs “B0V06” with $A_\perp = 0$ to isolate the contribution of longitudinal acoustic waves to coronal heating via mode-conversion. Set $A_\perp = 0$ and drive only $\delta v_\parallel$ at the 5-min p-mode band. This probes the chromospheric Alfvén generation rate via the $\partial_r \ln p$ coupling term (§B2 Elsässer source S_{refl}).

28.10 Per-star calibration scaling

For an arbitrary star, the solar calibration is scaled via Shimizu+22 Eq. 40:

$$\langle \delta v_0 \rangle \propto (T_{\text{eff}}^4 / \rho_0)^{1/3}, \quad \omega_{\text{max}}^{-1} \propto c_{s,0} R_*^2 / M_* \text{ (photospheric transit),}$$

with solar anchors $\langle \delta v_0 \rangle_\odot = 1.25 \text{ km/s}$, $(\omega_{\text{max}}^{-1})_\odot = 0.3 \text{ min}$. Numbers in Suzuki+25 Table 3:

- α Boo: $\langle \delta v_0 \rangle = 2.50 \text{ km/s}$, $\omega_{\text{max}}^{-1} = 150 \text{ min}$, $\omega_{\text{min}}^{-1} = 1.5 \times 10^4 \text{ min}$.
- α Tau: $\langle \delta v_0 \rangle = 2.56 \text{ km/s}$, $\omega_{\text{max}}^{-1} = 340 \text{ min}$, $\omega_{\text{min}}^{-1} = 3.4 \times 10^4 \text{ min}$.

28.11 □ Verification checkpoints

- tests/test_athena_mhd_driver_spectrum.cu — build the driver time-series over $10^6 f_{\text{min}}^{-1}$, compute the FFT-measured spectrum, assert $\log P$ vs $\log f$ has slope -1 ± 0.05 within the driven band.
- tests/test_athena_mhd_driver_variance.cu — compute $\langle [z^+]^2 \rangle$ over a multi-period sample; assert within 5% of target $\langle \delta v^2 \rangle$.
- tests/test_athena_mhd_absorbing_bc.cu — inject a Gaussian down-going Alfvén pulse, measure the reflected-amplitude coefficient; lock $R < 10^{-4}$ (WKB floor).
- tests/test_athena_mhd_driver_reproducibility.cu — same seed produces byte-identical driver output across runs; reseed gives uncorrelated phases.

The absorbing-BC test is the non-trivial one: a wrong ghost-extrapolation order shows up here as a visible reflection pulse but would pass every other spectrum / variance test silently.

29 F1. Oblique linear MHD wave: rotated eigenvectors

sympy script: `scripts/f1_oblique_linwave.py` **verified:** spectrum invariance under $B_y \leftrightarrow B_z$ split when $|B|$, $B \cdot \hat{\mathbf{k}}$, ρ , p are held fixed (20 random trials, max err 4.9×10^{-15}); solenoidal constraint $\mathbf{k} \cdot \delta \mathbf{B} = 0$ for rotated eigenvector; c_f^2 reduces to §A3 form at $\theta = 0$. **code checkpoints:** `AthenaMHDSolver::init_linear_wave_oblique` (to be added); `tests/test_athena_mhd_linwave_oblique.cu` (A1 test).

29.1 Motivation

The §A3 MHD eigensystem is derived for 1D propagation along $\hat{\mathbf{x}}$. For a 2D convergence test with wave-vector $\mathbf{k} = (k_x, k_y)$ — Stone+08 §6.2 uses $\mathbf{k} \cdot \mathbf{L} = (2, 1)$ giving $\theta = \arctan 2 \approx 63.4^\circ$ on a 2×1 domain — we need the eigenvector *rotated* to the oblique direction.

This is the natural Phase A1 test: if the 1D linwave convergence (§A11) passes at $p \approx 2$ but the 2D oblique test fails, the bug is **purely in the $\mathbf{x} / \mathbf{y}$ flux coupling** (i.e., the VL2 corrector wraps or the CT corner-EMF averaging). No such bug is caught by 1D-only tests.

29.2 Rotation rule for primitive-form eigenvector

The primitive 7-vector eigenvector in the $\hat{\mathbf{x}}$ -frame from §A3:

$$\mathbf{r} = (\delta\rho, \delta v_x, \delta v_y, \delta v_z, \delta B_y, \delta B_z, \delta p)^T$$

(with $\delta B_x = 0$: the B_x component is not a wave variable in 1D, see §A3 discussion).

In the rotated frame with $\hat{\mathbf{k}} = (\cos \theta, \sin \theta, 0)$, the vector components transform under $R(\theta)$:

$$\begin{pmatrix} \delta v'_x \\ \delta v'_y \end{pmatrix} = R(\theta) \begin{pmatrix} \delta v_x \\ \delta v_y \end{pmatrix}, \quad \begin{pmatrix} \delta B'_x \\ \delta B'_y \end{pmatrix} = R(\theta) \begin{pmatrix} 0 \\ \delta B_y \end{pmatrix}. \quad (\text{F1-rotation})$$

$\delta\rho, \delta v_z, \delta B_z, \delta p$ are scalars — invariant under the z-axis rotation. Explicitly:

$$\delta B'_x = -\sin \theta \delta B_y, \quad \delta B'_y = +\cos \theta \delta B_y. \quad (\text{F1-B-rotation})$$

29.3 Solenoidal constraint check

The rotated $\delta \mathbf{B}$ must still satisfy $\nabla \cdot \delta \mathbf{B} = 0$, i.e., for a plane wave $\mathbf{k} \cdot \delta \mathbf{B} = 0$.

$$\mathbf{k} \cdot \delta \mathbf{B} = k_0 (\cos \theta \cdot (-\sin \theta \delta B_y) + \sin \theta \cdot \cos \theta \delta B_y) = 0.$$

Sympy-verified symbolically. This is the fundamental reason the rotation works: $\delta \mathbf{B}$ in the unrotated frame is perpendicular to $\hat{\mathbf{x}}$ (via $\delta B_x = 0$), and the rotation preserves orthogonality to the rotated axis.

29.4 Wave-speed formula in the oblique frame

Under rotation, c_{Ax} in the §A3 formula must be replaced by the Alfvén speed component along the wave direction:

$$c_{A,k} \equiv (\mathbf{B} \cdot \hat{\mathbf{k}}) / \sqrt{\rho}. \quad (\text{F1-cAk})$$

Then the fast-magnetosonic speed remains:

$$c_f^2 = \frac{1}{2}[(c_{s_0}^2 + c_A^2) + \sqrt{(c_{s_0}^2 + c_A^2)^2 - 4 c_{s_0}^2 c_{A,k}^2}],$$

with $c_A^2 = |B|^2 / \rho$ unchanged (it is the *total* Alfvén speed, not projected). Sympy verification: setting $\theta = 0$ (unrotated frame) reduces (F1-cAk) to $c_{Ax} = B_x / \sqrt{\rho}$ as in §A3.

29.5 Numerical verification

Random 20-state numerical check: given $(\rho, p, |\mathbf{B}|, B \cdot \hat{\mathbf{k}})$ fixed, the 7-wave spectrum is identical for *any* decomposition of $\mathbf{B}$ into B_y, B_z components. This confirms the spectrum depends only on the invariants $(|B|, B \cdot \hat{\mathbf{k}}, \rho, p)$ — equivalently, it is rotationally invariant around $\hat{\mathbf{k}}$.

Max eigenvalue error over 20 trials: 4.9×10^{-15} .

29.6 Stone+08 §6.2 oblique-convergence setup

1. Domain $L_x = 2, L_y = 1$, fully periodic.
2. Wave vector $\mathbf{k} = 2\pi(1, 2)/L$ pointed diagonally; the wave crosses the domain in one period.
3. Background state from §A11: $\rho_0 = 1, p_0 = 1/\gamma, \mathbf{v}_0 = 0, \mathbf{B}_0 = (1, \sqrt{2}, 1/2), \gamma = 5/3$.
4. For each of the 4 modes (fast, Alfvén, slow, entropy):
 - Compute unrotated eigenvector $\mathbf{r}$ from §A3.
 - Compute θ from (k_x, k_y) ; rotate vector components of $\mathbf{r}$ per (F1-rotation).
 - Plant IC on the 2D grid: $\mathbf{W}(x, y, 0) = \mathbf{W}_0 + A \mathbf{r}_{\text{rotated}} \cos(\mathbf{k} \cdot \mathbf{x})$ with $A = 10^{-6}$.
5. Evolve for one wave period $T = 2\pi/(\lambda |\mathbf{k}|)$ where λ is the oblique wave speed.
6. Measure L^1 error: $\varepsilon_{L^1}(N) = \frac{1}{N_x N_y} \sum |\mathbf{W}^{n+1} - \mathbf{W}^0|$.

29.7 Pass criteria (A1 test)

1. For each of 4 modes $\times$ 3 resolutions ($N = 32, 64, 128$), $\varepsilon_{L^1}(N) \propto N^{-p}$ with $p \geq 1.8$.
2. $\max_t |\nabla \cdot \mathbf{B}| < 10^{-10}$ throughout (CT lock under 2D oblique propagation, the stronger case than 1D).
3. Solver remains stable for all modes (entropy / Alfvén / fast / slow must all propagate without NaN for $N \in \{32, 64, 128\}$).

29.8 Why this specifically tests 2D coupling

A bug localised to x-sweep or y-sweep alone will NOT manifest in 1D linwave tests (§A11) because only one direction is exercised. It WILL show up here if:

- **CT corner-EMF averaging** (GS05 §A5) has a wrong 4-point weight — the rotated δB_y requires exact y-flux of E_z^x and x-flux of E_z^y contributions to cancel.

- **VL2 predictor-corrector** has mismatched $dt/2$ vs dt between directions — the oblique wave accumulates directional phase error.
- **PLM slope limiter** has different logic in x vs y — the oblique wave tests both slopes simultaneously, while §A11 tests only one.

29.9 □ Verification checkpoints

- tests/test_athena_mhd_linwave_oblique.cu — Phase A1 test, 4 modes $\times$ 3 resolutions, locks the three pass criteria above.

Failure on A1 after §A11 passes indicates a specifically 2D-coupling bug — isolate by rerunning 1D (§A11), 2D-aligned (rerun §A11 on $L_x = 1, L_y = 1$ rotated 0°), 2D-oblique (this test).

30 F2. 2D MHD turbulence spectrum and ν_{eff} extraction

sympy script: scripts/f2_mhd_turbulence_spectrum.py **verified:** K41 log-log slope $-5/3$; IK65 slope $-3/2$; dissipation cutoff $k_{\text{diss}} = (\epsilon/\nu_{\text{eff}}^3)^{1/4}$; inversion $\nu_{\text{eff}} = (\epsilon/k_{\text{diss}}^4)^{1/3}$; scheme-order scaling $k_{\text{diss}}(N) \propto N^{3/2}$ for $\nu_{\text{eff}} \propto h^2$. **code checkpoints:** scripts/analyze_orszag_tang_spectrum.py (driver-external analysis); docs/projects/mhd_verification/phase_A_results.md — table entries.

30.1 Motivation

The Orszag-Tang (OT) vortex, run to $t = 0.5$, develops a fully nonlinear MHD turbulent cascade. Its energy spectrum $E(k)$ provides the **direct, resolution-independent** measurement of our solver’s effective viscosity ν_{eff} — the unique number that determines whether our 2D MHD turbulence runs are resolving the inertial range of interest.

Without this derivation, the A2 analysis script has no basis to interpret the spectrum cutoff or claim a quantitative ν_{eff} .

30.2 Kolmogorov (K41) and Iroshnikov-Kraichnan (IK65)

Two competing predictions for the 2D MHD inertial-range spectrum:

$$\boxed{E_K(k) = C_K \epsilon^{2/3} k^{-5/3}, \quad E_{IK}(k) = C_{IK} (\epsilon v_A)^{1/2} k^{-3/2}.} \quad (\text{F2-K41, F2-IK})$$

The slopes differ: - **K41**: hydrodynamic Kolmogorov-Obukhov cascade, applies when kinetic and magnetic energies are approximately equipartitioned and the cascade is local in k -space. - **IK65**: Iroshnikov-Kraichnan, applies when strong Alfvén-wave collisions dominate the cascade; the extra factor $v_A^{1/2}$ encodes the Alfvén-wave crossing time.

Both slopes are sympy-verified via symbolic log differentiation.

For OT at $t = 0.5$ the literature consensus (Dahlburg-Picone 1989, Politano-Pouquet 1989, Biskamp-Welter 1989) places the **observed** slope between $-5/3$ and $-3/2$, closer to the K41 value.

30.3 Dissipation cutoff and ν_{eff} inversion

Below the viscous scale, the cascade is truncated by dissipation. Classical Kolmogorov dissipation scale:

$$k_{\text{diss}} = (\epsilon/\nu_{\text{eff}}^3)^{1/4}. \quad (\text{F2-kdiss})$$

Inverting, given a *measured* k_{diss} from the spectrum:

$$\boxed{\nu_{\text{eff}} = (\epsilon/k_{\text{diss}}^4)^{1/3}}. \quad (\text{F2-nu-inv})$$

Sympy-verified: (F2-kdiss) and (F2-nu-inv) are each other's functional inverse.

30.4 Scheme-order consistency

For a 2nd-order finite-volume scheme with $\nu_{\text{eff}} \propto \Delta x^2$:

$$\nu_{\text{eff}}(N) = C_{\text{visc}}/N^2, \quad k_{\text{diss}}(N) = (\epsilon/C_{\text{visc}}^3)^{1/4} N^{3/2}. \quad (\text{F2-scaling})$$

Sympy-verified: $d \log k_{\text{diss}}/d \log N = 3/2$.

Doubling the resolution should shift k_{diss} by a factor $2^{3/2} = 2.83$. This is the A2 consistency check: $k_{\text{diss}}(256)/k_{\text{diss}}(128) \in [2.0, 3.5]$.

30.5 Measurement protocol (A2 test)

1. Run OT at $N \in \{128, 256, 512\}$ with `init_orzag_tang()` to $t = 0.5$.
2. Dump VTK of (v_x, v_y, B_x, B_y) at $t = 0.5$.
3. FFT each to 2D (k_x, k_y) ; compute 1D axisymmetric spectrum via shell averaging:

$$E(k) = \frac{1}{2} \sum_{k-1/2 < |\mathbf{k}'| \leq k+1/2} (|\hat{v}|^2 + |\hat{B}|^2).$$

4. Identify inertial range: sliding window of log-log slope; pick k_{iner} where slope is stable over 1 decade.
5. Identify dissipation cutoff: smallest k where $E(k)$ drops to $< 10^{-3}$ of its inertial-range peak (or use the break in slope from $-5/3$ to exponential fall).
6. Compute ν_{eff} via (F2-nu-inv) using measured ϵ (energy flux from $-dE_{\text{tot}}/dt$ at $t = 0.5$) and k_{diss} .
7. Verify scheme-order scaling (F2-scaling) between $N=128 \rightarrow 256$ and $N=256 \rightarrow 512$.

30.6 Pass criteria (A2 test)

1. **Inertial-range slope at N=256:** fitted log-log slope in $[-1.8, -1.4]$ over a continuous decade in k . Slope outside this range = wrong physics or cascaded too weakly.
2. **k_{diss} scaling:** $k_{\text{diss}}(256)/k_{\text{diss}}(128) \in [2.0, 3.5]$ and $k_{\text{diss}}(512)/k_{\text{diss}}(256) \in [2.0, 3.5]$.
3. **$\nu_{\text{eff}}(N)$ table entry** written to the Phase A results document (no pass/fail judgement; the number itself is the deliverable).

30.7 Why this matters for later Suzuki physics

Estimated at our expected parameters ($\epsilon \sim 0.1$, $\nu_{\text{eff}} \sim 10^{-4}$ at $N = 128$, extrapolated to $\sim 10^{-5}$ at $N = 512$):

N	ν_{eff}	Re_{num}	inertial range (decades)
128	$\sim 10^{-4}$	$\sim 10^4$	≈ 1.2
256	$\sim 2 \times 10^{-5}$	$\sim 3 \times 10^4$	≈ 1.6
512	$\sim 6 \times 10^{-6}$	$\sim 10^5$	≈ 2.0

A Suzuki-type 2D Alfvén-turbulence extension needs $\text{Re}_m \gtrsim 10^3$ to set up a realistic inertial range. **At N=256 or above**, this condition is met. Below N=256 the cascade is marginally resolved; above N=512 we are firmly in the inertial regime.

This is the single most important quantitative output of Phase A.

30.8 □ Verification checkpoints

- `scripts/analyze_orszag_tang_spectrum.py` implements the protocol above and writes to `phase_A_results.md`.
- Solver correctness is already verified by the existing `test_athena_mhd_benchmarks.cu::test_orszag_tang_spectrum` (smoke + $\nabla \cdot \mathbf{B}$ lock); A2 is an **analysis**, not a new solver test.

If the spectrum violates pass criterion (1), the likely cause is a broken HLLD branch at low- β (OT has $\beta \sim 0.01$ late in time); if (2) fails, the effective viscosity doesn't scale like a 2nd-order scheme, indicating a time-stepping bug (check VL2 corrector wraps in A8).

31 F3. CT round-off accumulation + B_{cc} aliasing in long-time field-loop

sympy script: `scripts/f3_ct_roundoff_and_bcc_aliasing.py` **verified:** random-walk round-off bound = worst-case / $\sqrt{n_{\text{step}}}$; midpoint reconstruction $B_{\text{cc}} = B + (h^2/8)B'' + O(h^4)$ for smooth B ; aliasing bound $|\Delta \text{ME}_{\text{cc}}| \leq C_{\text{alias}} A_0^2 \pi R (h/R)$, first-order in h , non-monotonic in t . **code checkpoints:** `tests/test_athena_mhd_field_loop_long.cu` — L1 (divB round-off), L3 (ME_{cc} aliasing bound).

31.1 Motivation

Phase A3 of the Phase A verification plan asks: **does CT actually preserve $\nabla \cdot \mathbf{B} = 0$ to machine precision over 10^4 steps**, not just through 10^2 as the short test covers?

The telescoping identity §A5 is algebraically exact in real arithmetic. In *floating-point* arithmetic each face update introduces a ULP-sized error, and the accumulation bound must be derived before the long-time test has a quantitative pass criterion.

A second, subtler issue: the long-time test measures magnetic energy through the **cell-centred reconstruction** B_{cc} , not the face-stored B_f . This diagnostic aliases the C^0 kink at the field-loop boundary $r = R$ as the loop translates, producing oscillations in $\text{ME}_{\text{cc}}(t)$ that **do not** violate CT. Without derivation, the test could mistake diagnostic aliasing for a solver bug.

31.2 Q1: CT round-off accumulation bound

One CT face update:

$$(B_x)_{i+1/2,j}^{n+1} = (B_x)_{i+1/2,j}^n - \frac{\Delta t}{\Delta y} (E_z^{i+1/2,j+1/2} - E_z^{i+1/2,j-1/2}).$$

In double precision (IEEE-754), each subtraction carries relative error $\leq \varepsilon_{\text{ULP}} = 2.22 \times 10^{-16}$. The corner- E_z contribution scales with $|\mathbf{B}|_\infty$ (HLLD flux scaling; consistency of the ideal-MHD Jacobian).

The discrete divergence of a cell is a **signed sum of 4 face values**:

$$(\nabla \cdot \mathbf{B})_{i,j} = \frac{B_{x,R} - B_{x,L}}{\Delta x} + \frac{B_{y,T} - B_{y,B}}{\Delta y}.$$

Per-step round-off residual:

$$|\Delta(\nabla \cdot \mathbf{B})|_{\text{per step}} \leq \frac{4\varepsilon_{\text{ULP}} |\mathbf{B}|_\infty}{h}. \quad (\text{F3-per-step})$$

Over n_{step} updates, two accumulation models:

$$\boxed{\max_t |\nabla \cdot \mathbf{B}| \leq \begin{cases} 4 n_{\text{step}} \varepsilon_{\text{ULP}} |\mathbf{B}|_\infty / h & (\text{coherent, worst case}) \\ 4 \sqrt{n_{\text{step}}} \varepsilon_{\text{ULP}} |\mathbf{B}|_\infty / h & (\text{random walk}) \end{cases}} \quad (\text{F3-bound})$$

Sympy-verified: the random-walk form equals the worst-case form divided by $\sqrt{n_{\text{step}}}$.

Numeric check for the A3 test parameters ($\varepsilon_{\text{ULP}} = 2.22 \times 10^{-16}$, $|\mathbf{B}|_\infty = 1$, $h = 1/128$, $n_{\text{step}} \approx 10^4$):

Bound	Value
Worst-case (F3-bound)	1.14×10^{-9}
Random-walk (F3-bound)	1.14×10^{-11}
Measured (A3 test)	1.85×10^{-15}

The measured value is **4 orders of magnitude tighter** than even the random-walk bound. Interpretation (Gardiner-Stone 2005 §3.4.1): the CT stencil is sign-symmetric in the 4 corner contributions to each cell, so round-off cancels to within 1 ULP rather than accumulating as random walk. CT is not just algebraically exact — it is *round-off exact* on realistic hardware.

The A3 test lock $\max |\nabla \cdot \mathbf{B}| < 1\text{e-}10$ uses a safety margin between the measured 10^{-15} and the worst-case bound 10^{-9} .

31.3 Q2: B_{cc} reconstruction aliasing at the $r = R$ kink

The diagnostic cell-centred B is computed by midpoint averaging:

$$(B_{x,\text{cc}})_{i,j} = \frac{1}{2}((B_{x,f})_{i-1/2,j} + (B_{x,f})_{i+1/2,j}).$$

Smooth B behaviour (sympy-verified):

$$B_{x,\text{cc}} = B_x(x_i) + \frac{h^2}{8}B_x''(x_i) + \mathcal{O}(h^4). \quad (\text{F3-midpoint})$$

So for a smooth field, $\text{ME}_{\text{cc}}(t) - \text{ME}_{\text{true}}(t) = \mathcal{O}(h^2)$ is a small, smooth, t -independent bias.

Field-loop IC pathology. The GS05 field-loop has $|\mathbf{B}|$ step-function at $r = R$:

$$|\mathbf{B}|(r) = \begin{cases} A_0 & r < R \\ 0 & r > R \end{cases}.$$

At the kink, B is C^0 but not C^1 . The Taylor expansion in (F3-midpoint) fails, and the midpoint reconstruction has local error $\mathcal{O}(h)$ — **a full order worse** than the smooth case.

As the loop translates with velocity $\mathbf{v} = (v_x, v_y)$, the kink aliases successively across different cell boundaries. Let $N_{\text{ring}} \approx 2\pi R/h$ be the number of cells crossed by the ring. The cell-integrated ME_{cc} picks up a phase-dependent $\mathcal{O}(A_0^2 \cdot h/R)$ aliasing bound:

$$\boxed{|\text{ME}_{\text{cc}}(t) - \text{ME}_{\text{cc}}(0)| \leq C_{\text{alias}} A_0^2 \pi R (h/R), \quad \text{oscillatory in } t.} \quad (\text{F3-aliasing})$$

Sympy-verified properties: - Quadratic in A_0 (quadratic in amplitude, as ME itself is). - Linear in h : halving the grid spacing halves the aliasing error.

Critical physical interpretation. CT conserves the *face-integrated flux* $\oint \mathbf{B} \cdot d\mathbf{S}$ through any closed discrete loop *exactly* — this is the content of §A5. The discrepancy $\text{ME}_{\text{cc}}(t) \neq \text{const}$ lives entirely in the **diagnostic reconstruction** $B \rightarrow B_{\text{cc}}$, not in the solver state. A test that measures “ME conservation” via B_{cc} will report $\mathcal{O}(1)$ oscillation in $\text{ME}_{\text{cc}}(t)$ on a translating field loop — this is **expected**, not a bug.

A3 test measurement ($N=128$, $A_0=1\text{e-}3$, $R=0.3$, 10 crossings, minmod limiter):

$$\text{ME}_{\text{cc}}(t=10)/\text{ME}_{\text{cc}}(0) \approx 1.56, \quad \max_t \text{ME}_{\text{cc}}/\text{ME}(0) \approx 1.56.$$

This is within the expected aliasing envelope (F3-aliasing) for the parameters: $h/R = 0.026$, $N_{\text{ring}} \approx 60$ cells, $C_{\text{alias}} \sim \mathcal{O}(10)$ after 600 sub-cell-boundary crossings.

31.4 Limiter sensitivity

The field-loop IC is more sensitive to the choice of slope limiter than smooth tests (e.g., linear-wave convergence). **Empirically:** - **van Leer harmonic** (limiter = 0): the loop is *unstable* over ~ 8 crossings; ME_{cc} grows by 10^4 from a compressible instability driven by the sharp kink. CT still preserves $\nabla \cdot \mathbf{B}$ to 10^{-14} even as this happens — a clean demonstration that CT constraint-preservation $\neq$ physical stability. - **minmod** (limiter = 1): the loop is stable over 10+ crossings; ME_{cc} oscillates bounded by (F3-aliasing).

The A3 test uses minmod for this reason. The loop-instability under van Leer is not a bug — it is the reason Stone+08 §6.3 switches to minmod for shock-containing problems and notes that field-loop tests specifically require a more dissipative limiter.

31.5 □ Verification checkpoints

Exactly what the A3 test tests/test_athena_mhd_field_loop_long.cu locks:

- **L1** $\max_t |\nabla \cdot \mathbf{B}| < 10^{-10}$ — 1 order below worst-case bound (F3-bound) and 4 orders above measured.
- **L3** $\max_t \text{ME}_{\text{cc}}/\text{ME}_0 < 3$ — well within envelope (F3-aliasing) for $A_0=1\text{e-}3$, $h=1/128$, $R=0.3$.
- **L4** $\text{ME}_{\text{cc}}(t_{\text{end}})/\text{ME}_0 > 0.5$ — solution is not decaying to zero (loop structure preserved).

Any of these failing flags a real bug (solver-level or IC-level, not aliasing).

32 F4. CPAW 2D long-time decay and η_{eff} extraction

sympy script: scripts/f4_cpaw_decay_eta_eff.py **verified:** ideal limit $\eta \rightarrow 0$ recovers $\omega = \pm v_A k$; weak- η expansion $\omega = v_A k - (i/2)\eta k^2 + \mathcal{O}(\eta^2)$; amplitude decay $\gamma = \eta k^2/2$; 2nd-order scaling $\gamma_{\text{num}}(N) \propto N^{-2}$ via modified-equation analysis; two-resolution inversion formula recovers scheme order q from γ_1, γ_2 .
code checkpoints: tests/test_athena_mhd_cpaw_longtime.cu — CPAW 2D at $N \in \{32, 64, 128\}$, measure amplitude decay, extract $\eta_{\text{eff}}(N)$.

32.1 Motivation

Phase A4 asks: **what is the numerical resistivity of athena_mhd on linear Alfvén waves?** This number is the single most important quantitative calibration for later physics work: every 2D Alfvén turbulence / Shimizu-style run must have physical damping rates $\gg \gamma_{\text{num}}(N)$ at the target resolution, or the reported dissipation is unphysical.

Before running the test we derive **three formulas** that the test post-processing uses:

1. The resistive dispersion relation for linearised Alfvén waves (established by field theory but rarely documented in one place).
2. The weak- η amplitude-decay rate $\gamma = \eta k^2/2$.
3. The scheme-order inversion formula $p = \log(\gamma_1/\gamma_2)/\log(N_2/N_1)$.

32.2 Q1: Resistive Alfvén dispersion

Linearised transverse MHD on a uniform background ($\rho_0, B_{r,0}, \mathbf{v}_0 = 0$) with finite resistivity η :

$$\partial_t v_{\perp} = (B_{r,0}/\rho_0) \partial_r B_{\perp},$$

$$\partial_t B_{\perp} = B_{r,0} \partial_r v_{\perp} + \eta \partial_r^2 B_{\perp}.$$

Plane-wave ansatz $(v_{\perp}, B_{\perp}) = (V, B) e^{i(kr - \omega t)}$ gives the 2×2 dispersion matrix. Setting its determinant to zero yields

$$\boxed{\omega^2 + i\eta k^2 \omega - v_A^2 k^2 = 0, \quad v_A \equiv B_{r,0}/\sqrt{\rho_0}.} \quad (\text{F4-disp})$$

Sympy-verified: the ideal limit $\eta \rightarrow 0$ recovers the Alfvén dispersion $\omega = \pm v_A k$.

32.3 Q2: Weak- η expansion and amplitude decay

Solving (F4-disp) and expanding around the outgoing branch $\omega_0 = v_A k$ to first order in η :

$$\boxed{\omega = v_A k - \frac{i}{2} \eta k^2 + \mathcal{O}(\eta^2), \quad \varepsilon_{\text{weak}} \equiv \eta k / v_A \ll 1.} \quad (\text{F4-weak})$$

Sympy-verified via `sp.series`.

Physical amplitude decay. With $e^{-i\omega t} = e^{-iv_A k t} \cdot e^{-(\eta k^2/2)t}$,

$$A(t) = A_0 \exp(-\gamma t), \quad \gamma = \frac{1}{2} \eta k^2. \quad (\text{F4-decay})$$

Both Elsässer modes $z^\pm$ decay at the same rate γ — resistivity is non-selective between counter-propagating Alfvén waves.

Factor of 1/2. Different conventions absorb or drop the 1/2; here we keep it explicit because the A4 test measures amplitudes (not energies) and converts measured $\gamma \rightarrow \eta$ via the inverse $\eta = 2\gamma/k^2$.

32.4 Q3: Numerical resistivity from modified-equation analysis

For a 2nd-order Godunov scheme (PLM + HLLD + VL2, see §A6–A8) on the linear Alfvén equation, standard modified-equation analysis (LeVeque 2002 §18) gives

$$\boxed{\eta_{\text{eff}}(h) = C_{\text{num}} h^2 v_A, \quad \gamma_{\text{num}}(N) = \frac{1}{2} C_{\text{num}} h^2 v_A k^2 \propto N^{-2}.} \quad (\text{F4-eta-eff})$$

C_{num} is a scheme-dependent $\mathcal{O}(1)$ constant that depends on limiter choice, HLLD branch, and CT corner-EMF weights. We do **not** compute C_{num} from first principles (it would require von-Neumann analysis of the full VL2 + HLLD + CT loop on the Alfvén mode — closed-form intractable); instead, we *measure* $\eta_{\text{eff}}(N)$ empirically and **check** the h^2 scaling.

Sympy-verified: $d \log \gamma_{\text{num}} / d \log h = 2$.

32.5 Q4: Two-resolution inversion for scheme order

Given measured decay rates at two resolutions:

$$\boxed{p = \frac{\log(\gamma_1/\gamma_2)}{\log(N_2/N_1)},} \quad (\text{F4-order})$$

for a scheme with true order q and $\gamma_i \propto N_i^{-q}$, this formula returns $p = q$ exactly. Sympy-verified.

For our 2nd-order solver we expect $p \approx 2$. Tolerances in the A4 test: $|p - 2| < 0.3$ on pairs $(32 \rightarrow 64), (64 \rightarrow 128)$.

32.6 Test-pass criteria (A4 CPAW long-time)

For $N \in \{32, 64, 128\}$, CPAW 2D (§B2, Tóth 2000 §3.2.2) travelling at $v_A = 1$ over 10 and 50 wave periods:

1. **Amplitude remains finite:** $A(t_{\text{end}})/A(0) \in [0.1, 1]$ for every N — no catastrophic blow-up, no complete dissipation.
2. **Monotone decay:** $A(t)$ monotone non-increasing (modulo small aliasing noise, similar to §F3 B_cc aliasing).
3. **Scheme-order consistency:** $|p_{32 \rightarrow 64} - 2| < 0.3$ AND $|p_{64 \rightarrow 128} - 2| < 0.3$.
4. η_{eff} **table entries** written to CSV for the Phase A results document:

N	γ_{num}	$\eta_{\text{eff}} = 2\gamma/k^2$	$\text{Re}_{\text{num}} = v_A L / \eta_{\text{eff}}$
32	(measured)	(derived via F4-decay)	(physical interpretation)
64	...	...	...
128	...	...	...

32.7 Why this matters for Suzuki physics

For $L = \sqrt{5}$, $v_A = 1$, $k_{\text{wave}} = 2\pi$ in our CPAW IC, the weak- η criterion $\eta k / v_A \ll 1$ gives $\eta \ll 0.16$. We expect measured $\eta_{\text{eff}}(128) \lesssim 10^{-4}$, corresponding to numerical magnetic Reynolds number $\text{Re}_m \sim 10^4$ on the domain scale.

When we extend to Suzuki-style problems (C8 chromosphere, E1 driver, §B1 flux tube), the physical magnetic Reynolds number at typical parameters is $\text{Re}_m^{\text{phys}} \sim 10^6\text{--}10^{10}$. **Our solver cannot resolve this directly;** physical η is irrelevant because $\text{Re}_m^{\text{num}} < \text{Re}_m^{\text{phys}}$. But for *turbulence* problems where the physical cascade only needs $\text{Re}_m \gtrsim 10^3$ to set up an inertial range, our solver at $N \geq 256$ is viable.

F4 gives us the quantitative version of that statement, per resolution.

32.8 □ Verification checkpoints

- tests/test_athena_mhd_cpaw_longtime.cu — will implement the A4 test following this derivation.

Pass criteria map directly from (F4-decay), (F4-eta-eff), (F4-order) to concrete numerical thresholds in the test file.